MYOCARDIAL INFARCTION

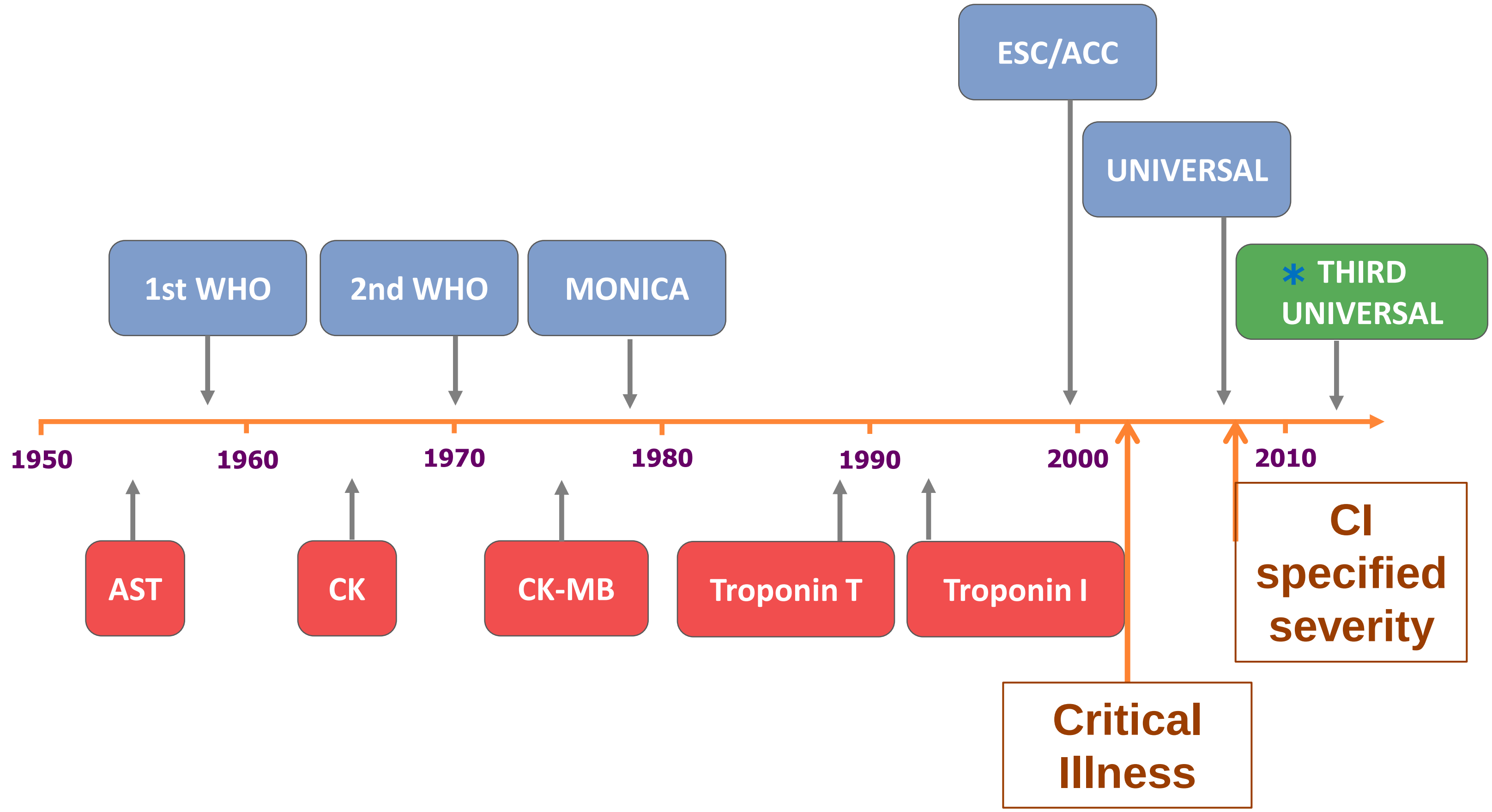
HEALTH CLAIMS FORUM

February 2015

Nick Boon
Cardiologist, Edinburgh
CMO, RGA

Reinsurance Group of America, Incorporated®

Changes in the Definition of Myocardial Infarction



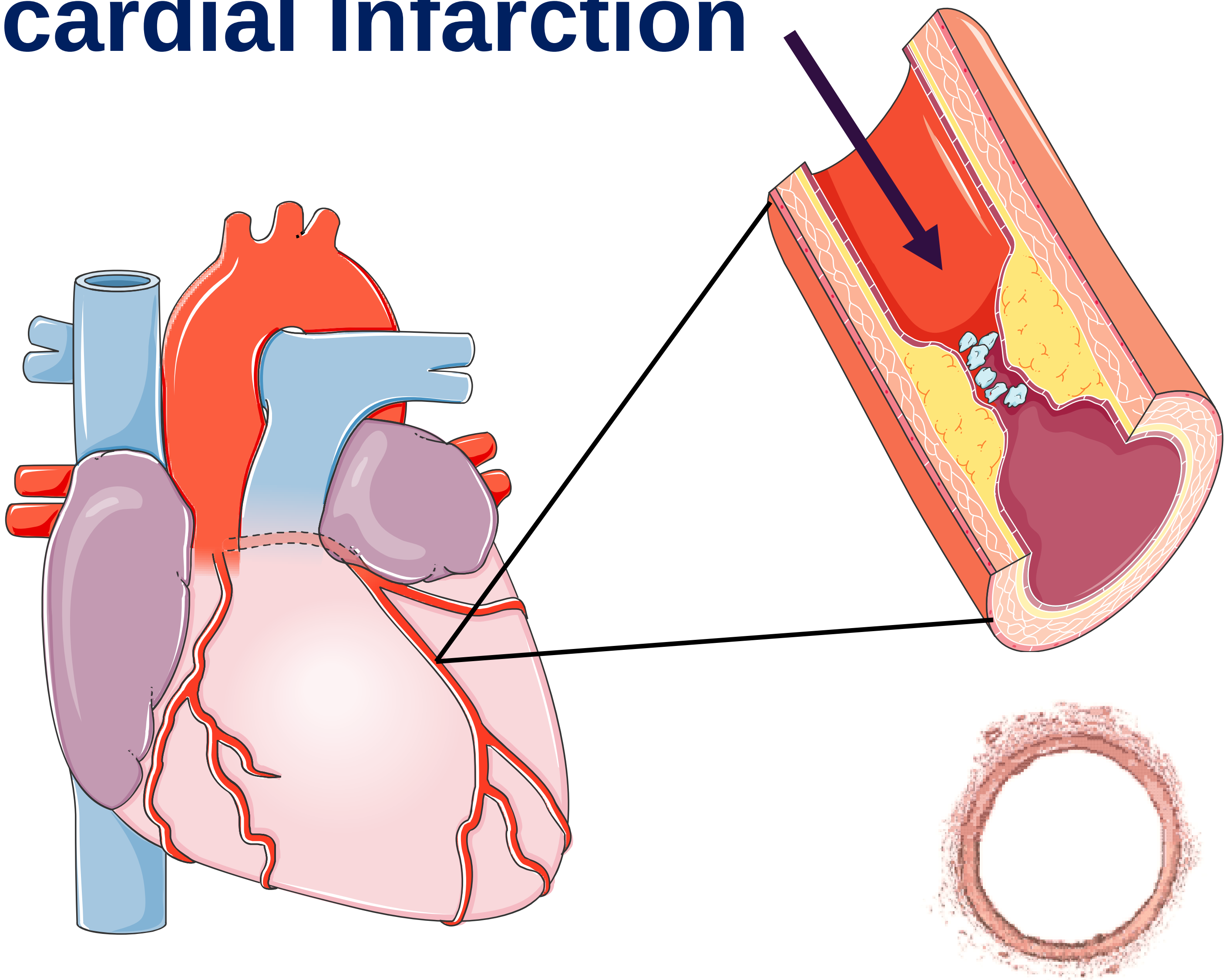
* *Eur Heart J. 2012 Oct;33(20):2551-2567.*

- **Background**
  - **- pathology**
  - **- diagnosis**
  - **- management**
  - **- definitions**

- **Real world experience**

- **Awkward claims**

Classical (type 1) Myocardial Infarction



3rd Universal Definition of MI

Type 1	Spontaneous myocardial infarction related to ischemia due to a primary coronary event such as plaque erosion and/or rupture, fissuring, or dissection
Type 2	Myocardial infarction secondary to ischemia due to either increased oxygen demand or decreased supply
Type 3	Sudden unexpected cardiac death often with symptoms suggestive of myocardial ischemia
Type 4	Associated with PCI/stent thrombosis
Type 5	Associated with CABG

http://circ.ahajournals.org/content/early/2012/08/23/CIR.0b013e31826e1058.citation

Type 2 Myocardial Infarction

Infarction due to increased myocardial oxygen demand, decreased supply or both

O_2 supply **O_2 demand**

Oxygenation
Coronary Flow
(tone and perfusion pressure)

Heart Rate
Blood Pressure
Contractility

Type 2 Myocardial Infarction

↑ O_2 demand

Tachycardia
Stress

↓ O_2 supply

Hypoxia
Shock

An Example of Type II Myocardial Infarction

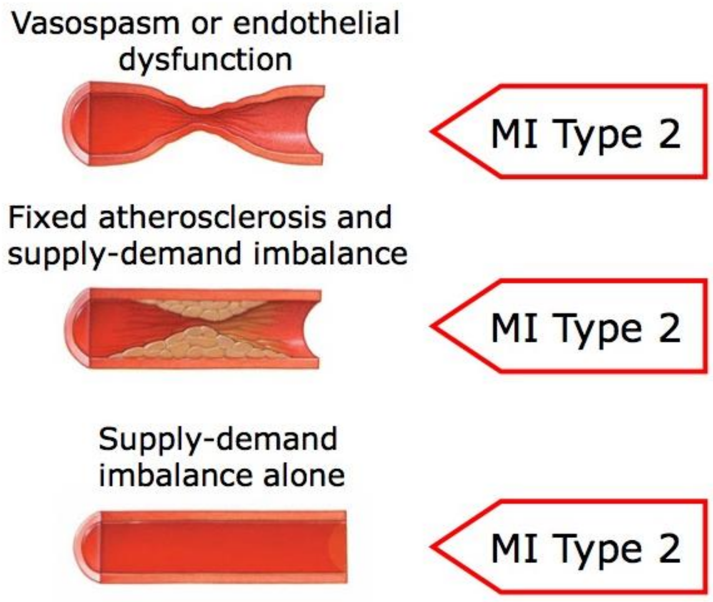
76 year old woman, occasional angina on effort
Short history of productive cough, intermittent chest pain

Temp 38.4O C, Pulse 114/min Bp 100/60 mm Hg
Oxygen Sats 78%
ECG - T wave inversion
CXR - lobar pneumonia
Troponin - 40 ng/l

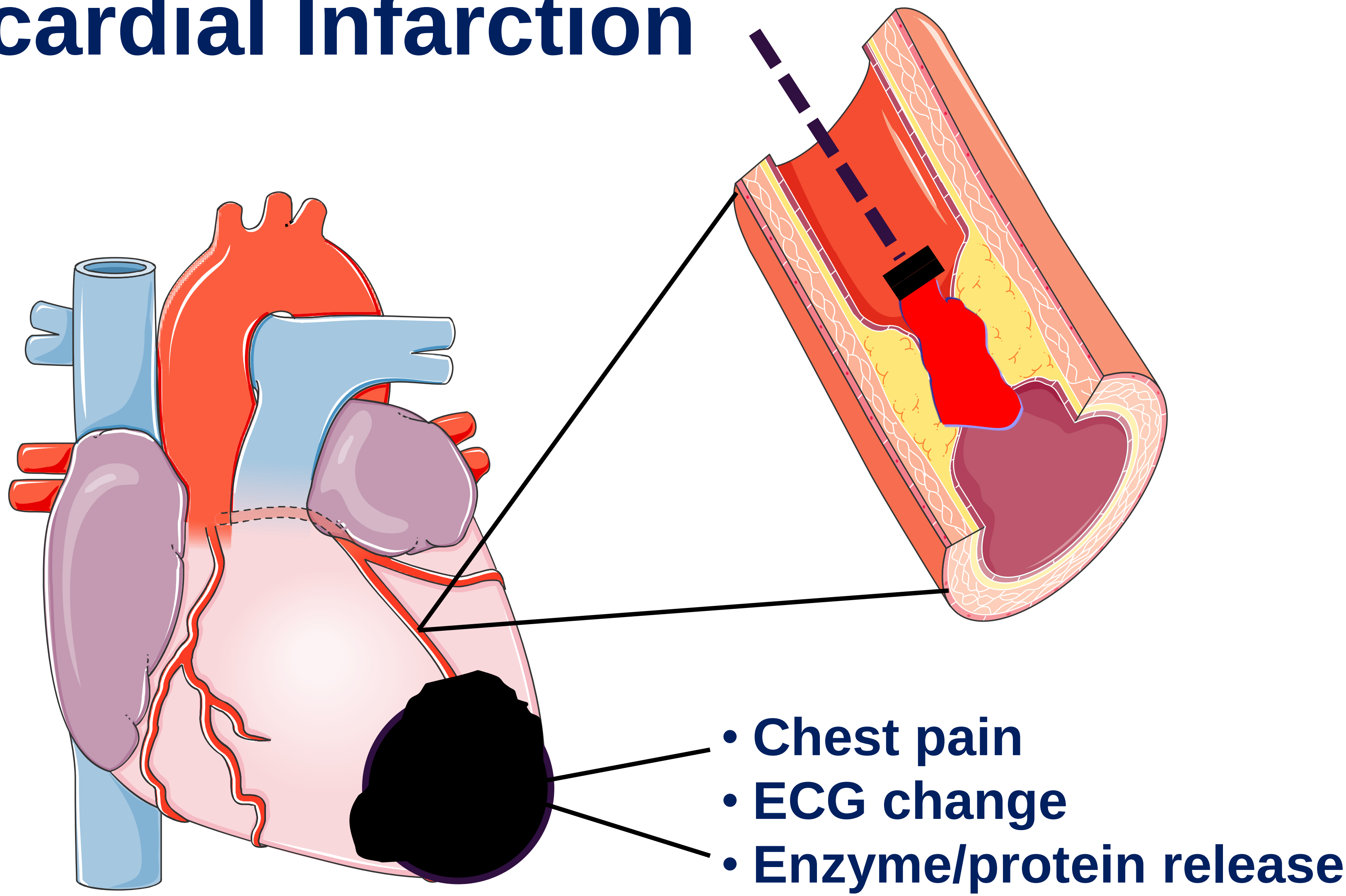
Angiogram – patent coronary arteries with moderately severe diffuse atheroma

Type 2 Myocardial Infarction

Angiography may show normal coronary arteries, mild, moderate or severe disease.



Cardinal Features of Myocardial Infarction

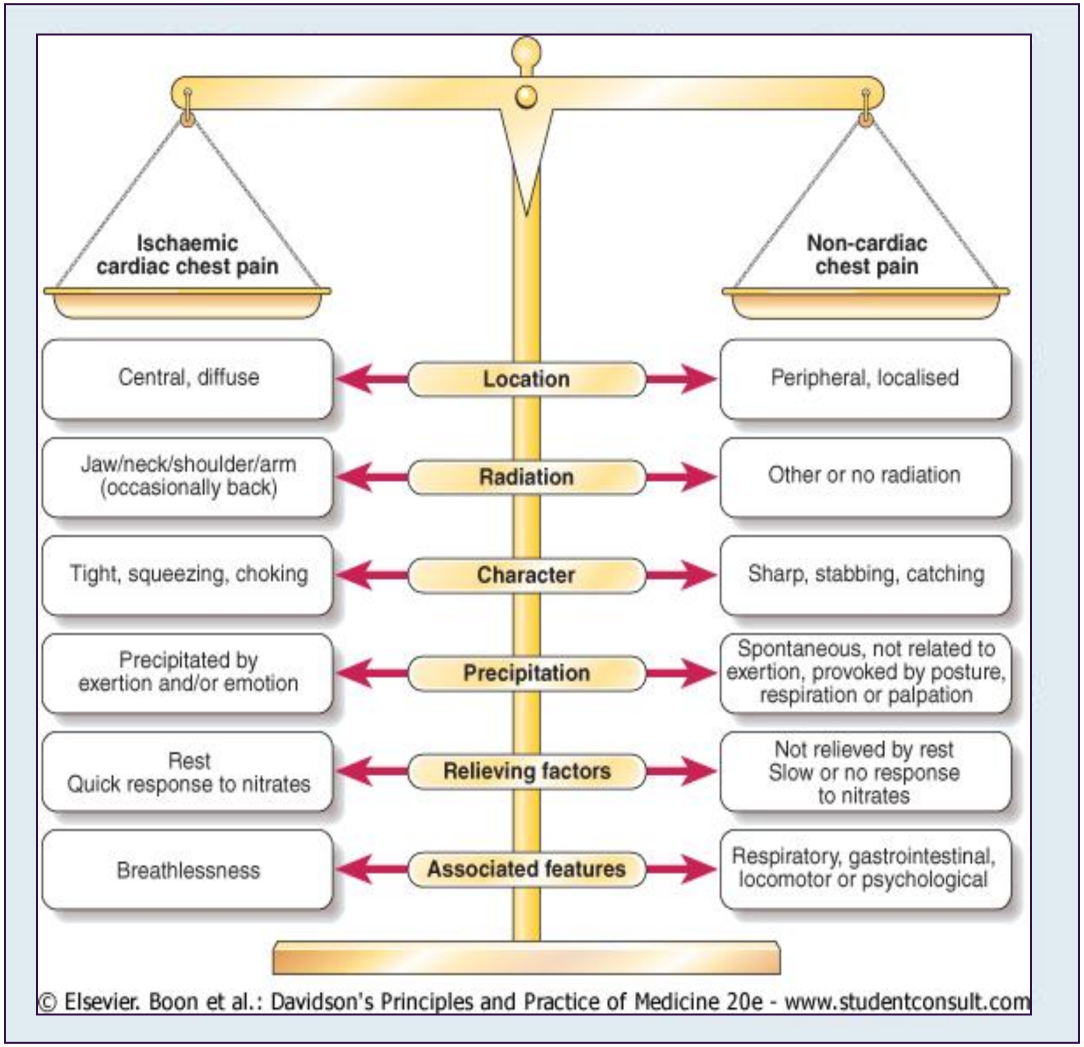


WHO criteria for the diagnosis of Myocardial Infarction

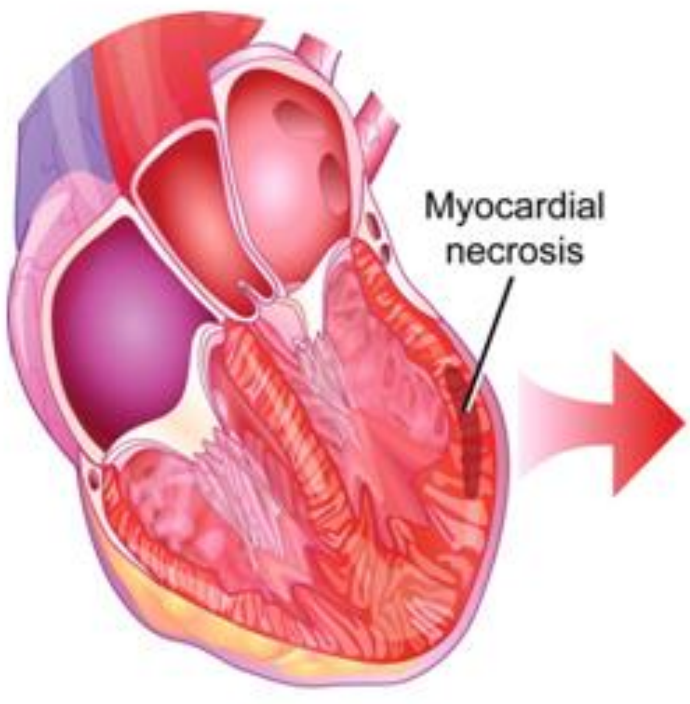
- Chest pain lasting more than 30 minutes
- Elevated (2 x normal) ‘cardiac enzymes’
- Typical ECG changes

1 criterion - possible MI
2 criteria - probable MI
3 criteria - definite MI

Cardiac Chest Pain



Biochemical Markers in Acute Myocardial Infarction



Biochemical Markers in Acute Myocardial Infarction - Specificity

Aspartate Transaminase (AST)

Liver, Lung, Skeletal muscle

Lactic Dehydrogenase (LDH)

Red blood cells, Liver, Lung, Skeletal muscle

Creatinine Kinase (CK)

Skeletal muscle, Brain, Kidney

CK MB isoenzyme

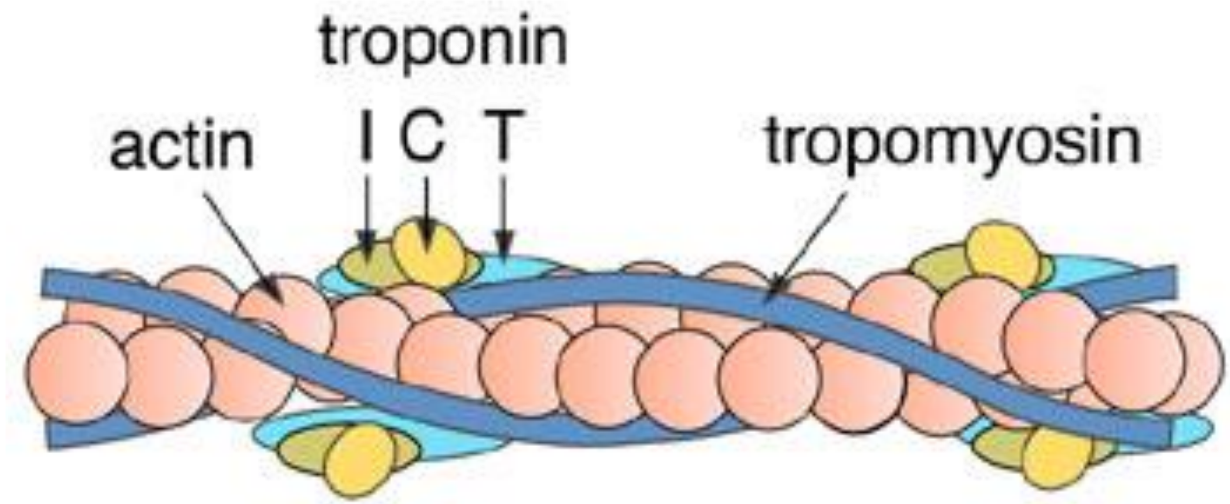
gut, tongue, diaphragm

Biochemical Markers in Acute Myocardial Infarction - Specificity

troponin is only found in myocytes

but

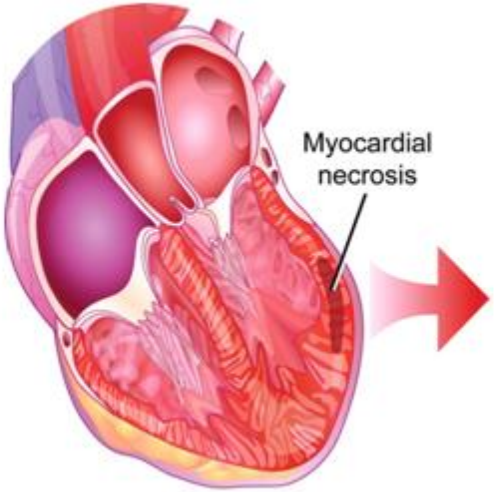
MI is NOT the only cause of troponin release

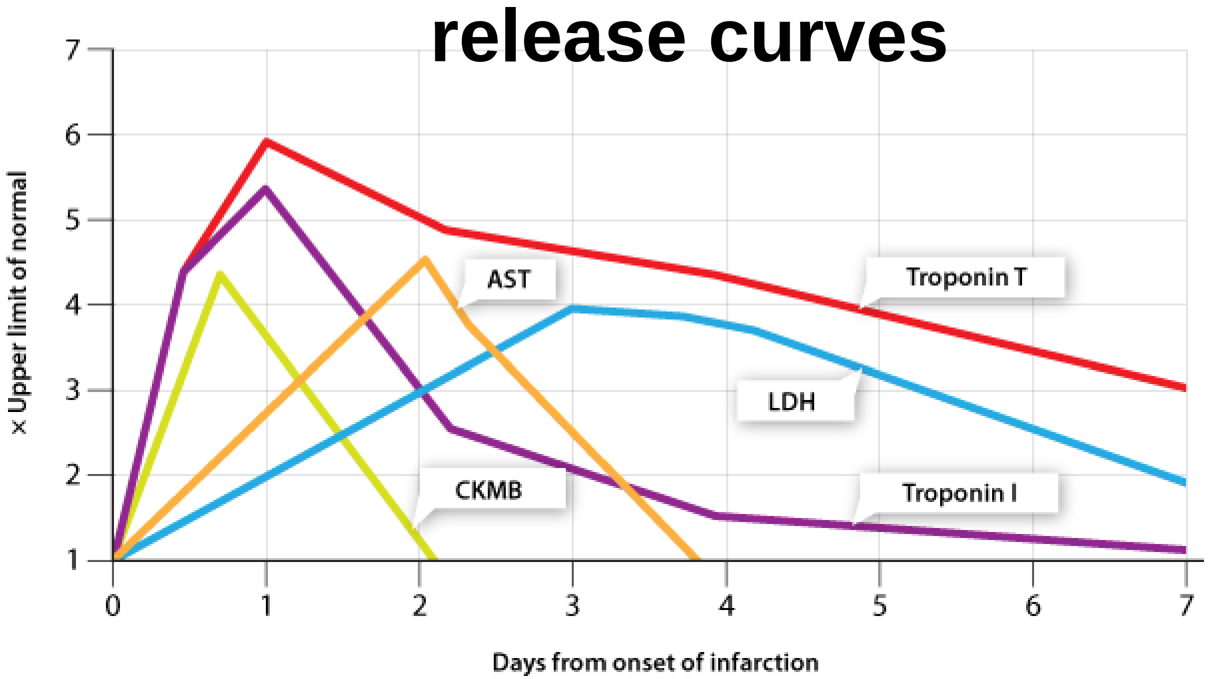


Non MI Causes of an Elevated Troponin

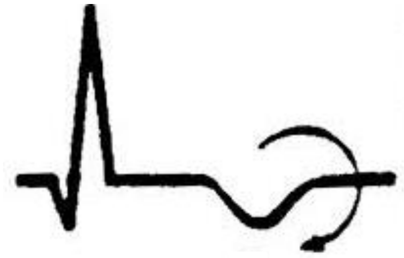
Apical ballooning syndrome
Heart failure
Pulmonary embolism
Myocarditis
Hypothyroidism
Renal failure
Acute neurological disorders (Stroke and SAH)
Critical illness (especially sepsis)
Burns
Drug toxicity

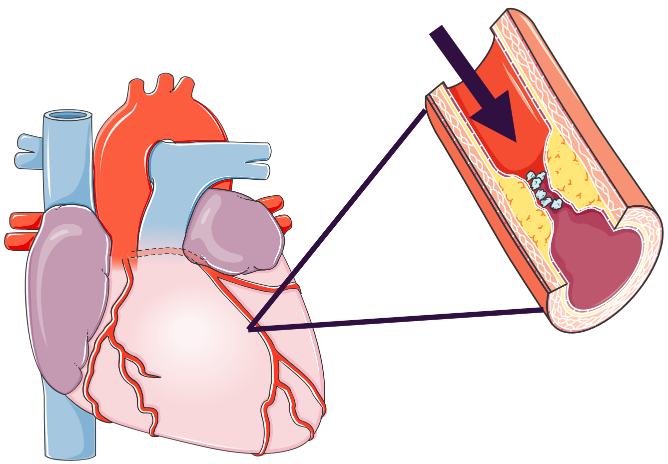
Biochemical Markers in Acute Myocardial Infarction





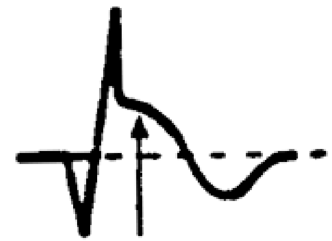
Non-ST Elevation

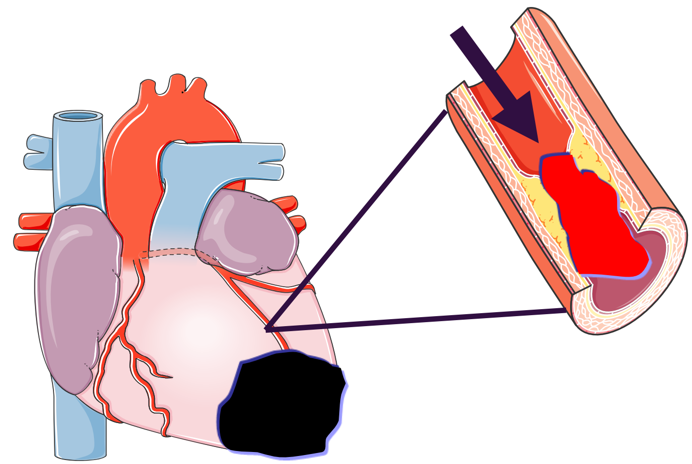




partial thickness (sub-endocardial) infarction

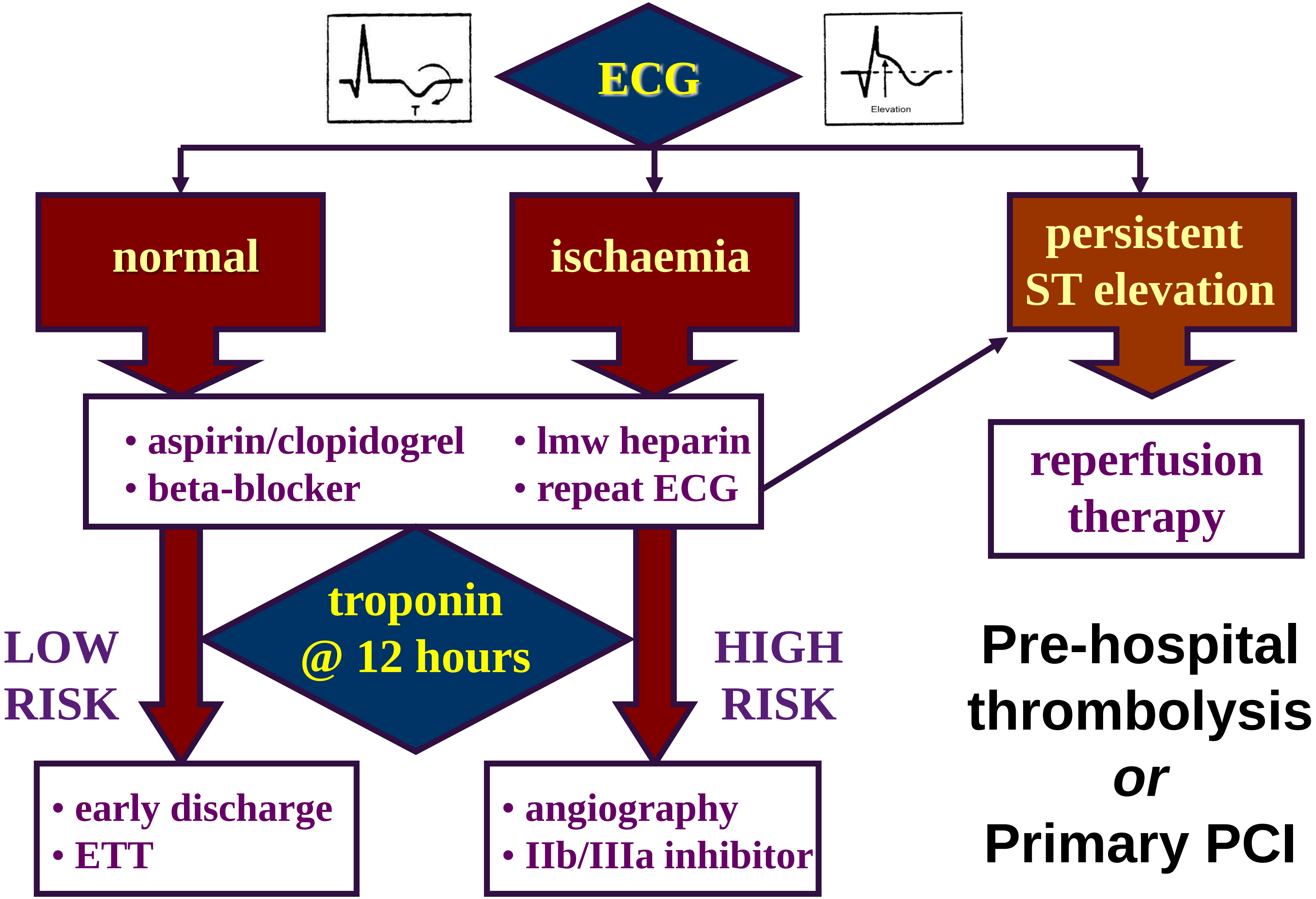
ST Elevation





evolving full-thickness infarction

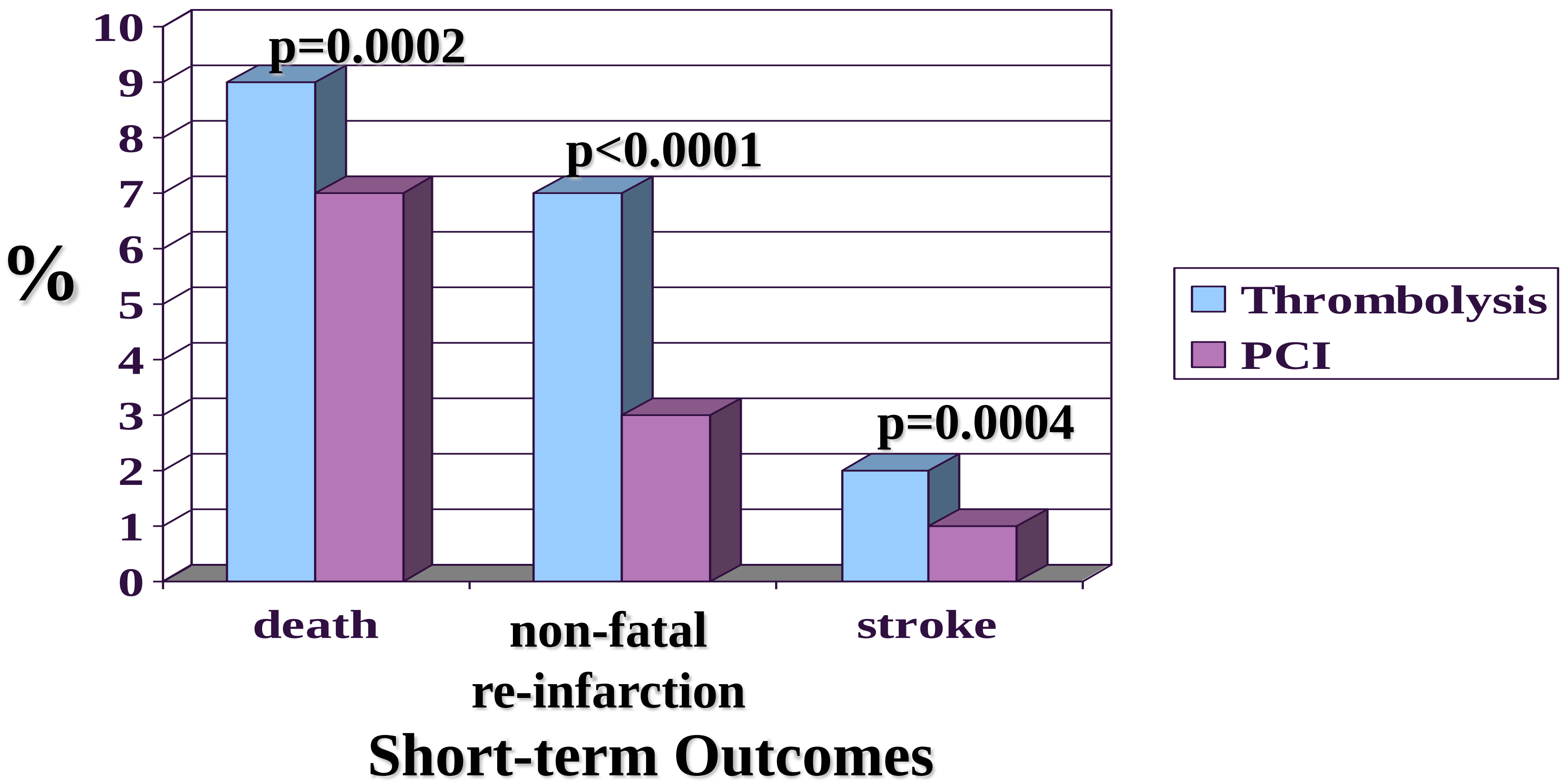
Early Management of an Acute Coronary Syndrome



Modified from ESC Guidelines

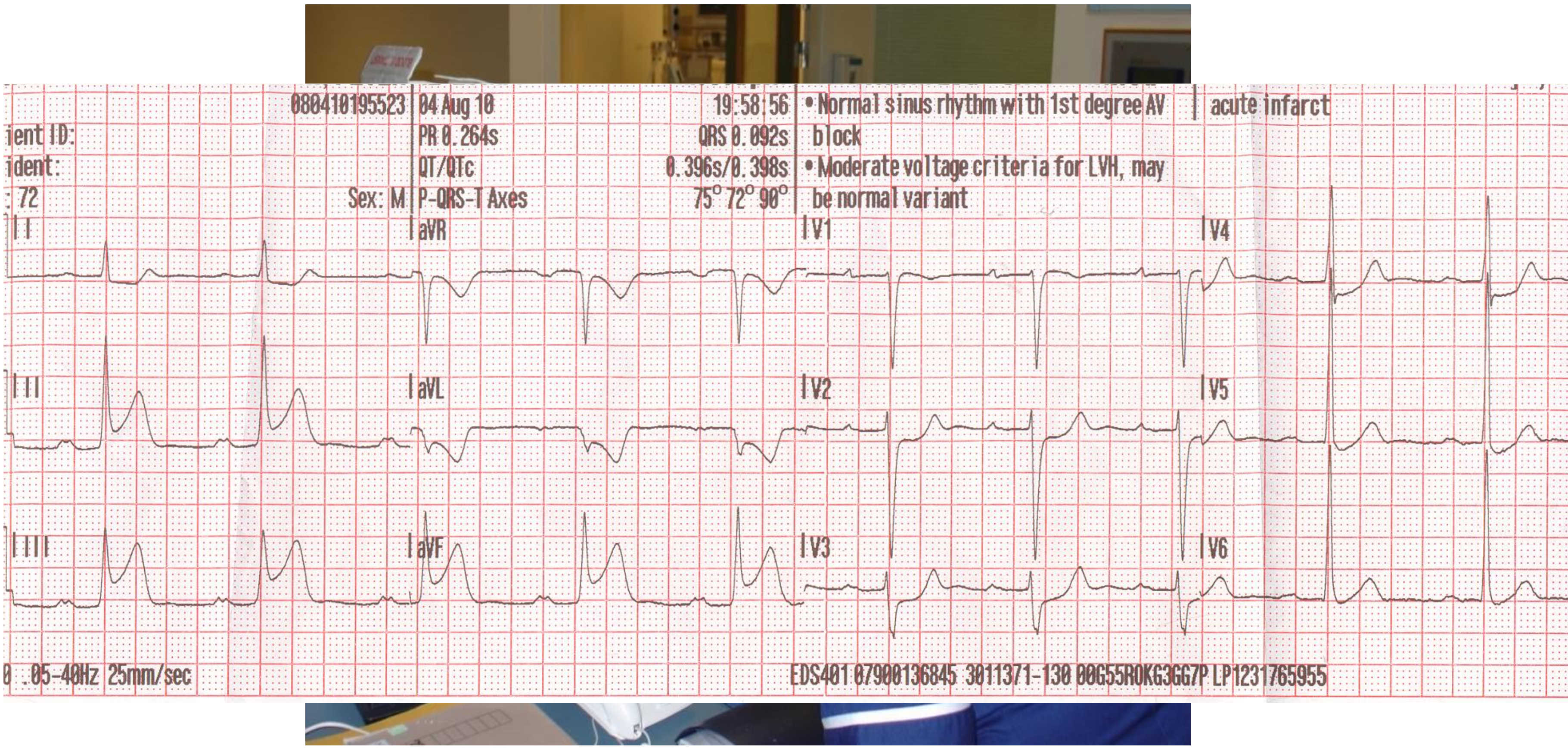
Primary PCI vs Hospital Thrombolysis meta-analysis of RCTs

23 trials : 7739 STEMI patients



Keeley et al. Lancet 2003; 361 : 12-20

23.56 hrs 14th August 2010



23.58 hrs 14th August 2010

72 year old man
Usually fit and well
Chest pain for 90minutes
7 minutes from RIE

00.06 hrs
t = 8 mins

LCA angio

00.16 hrs
t = 18 mins

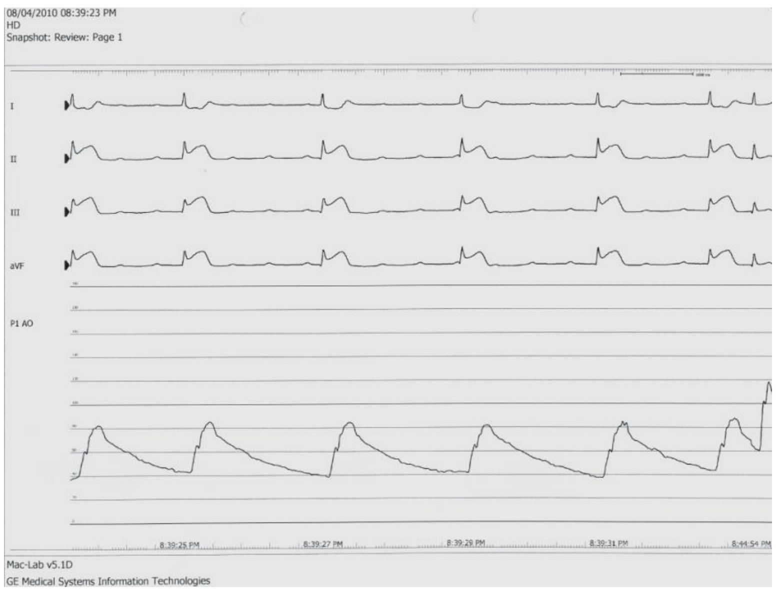
Heart Block,
Hypotension

RCA angio

00.17 hrs

t = 19 mins

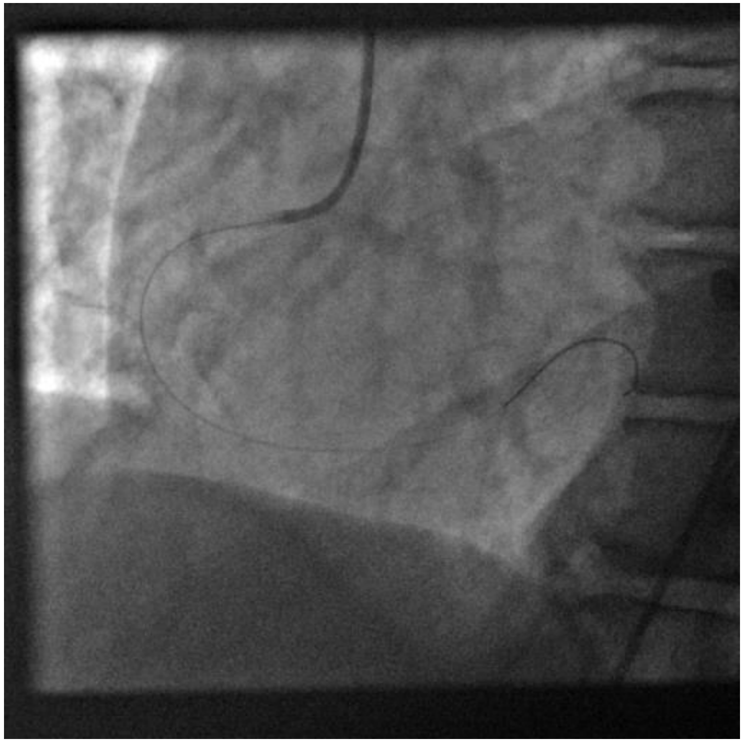
Heart Block, Hypotension

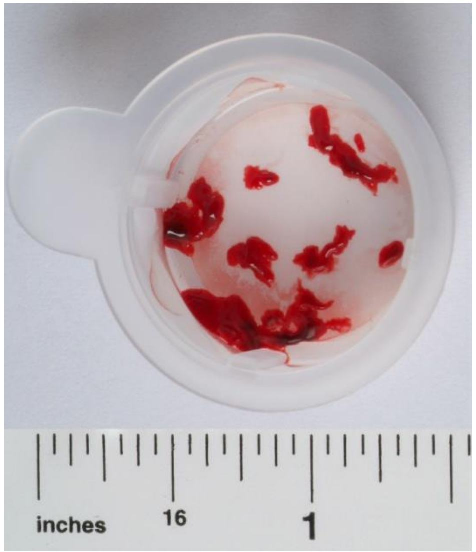


Thrombectomy

00.20 hrs

t = 22 mins



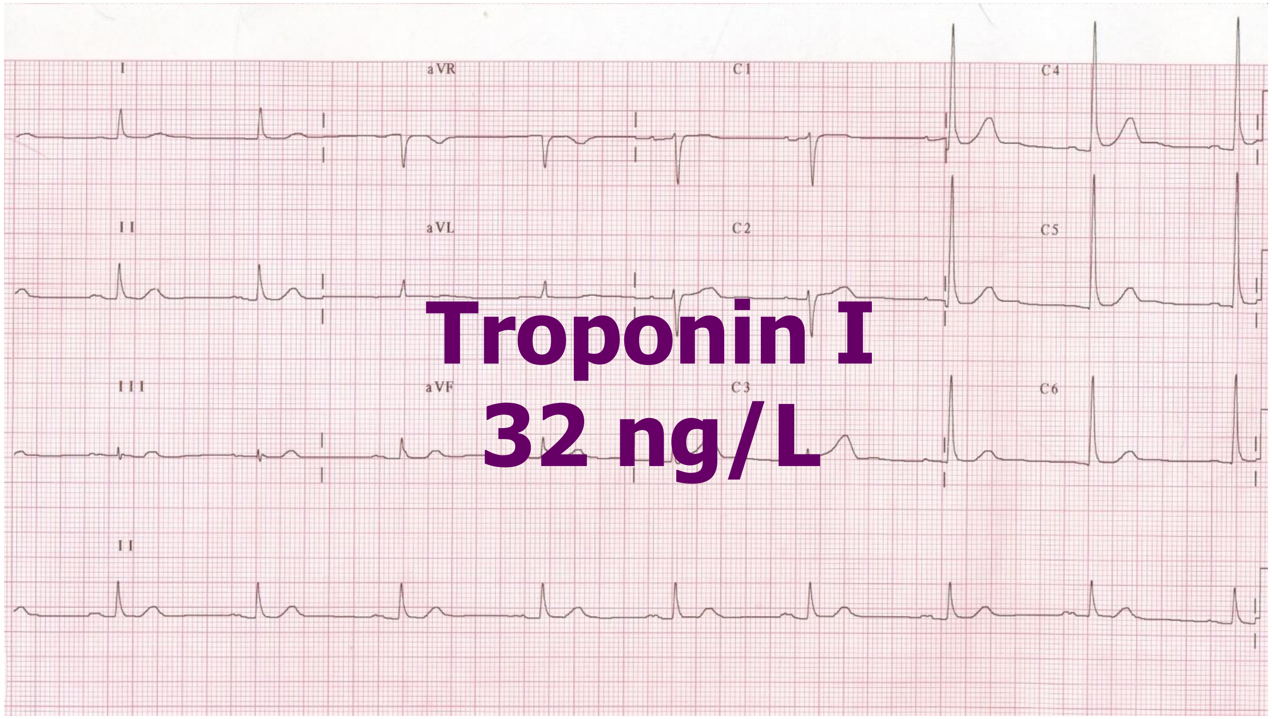


00.22 hrs
t = 24 mins

Pain free
HR 90/min
BP 130/80

PCI

01. 15 hrs
t = 77 mins



Troponin I
32 ng/L

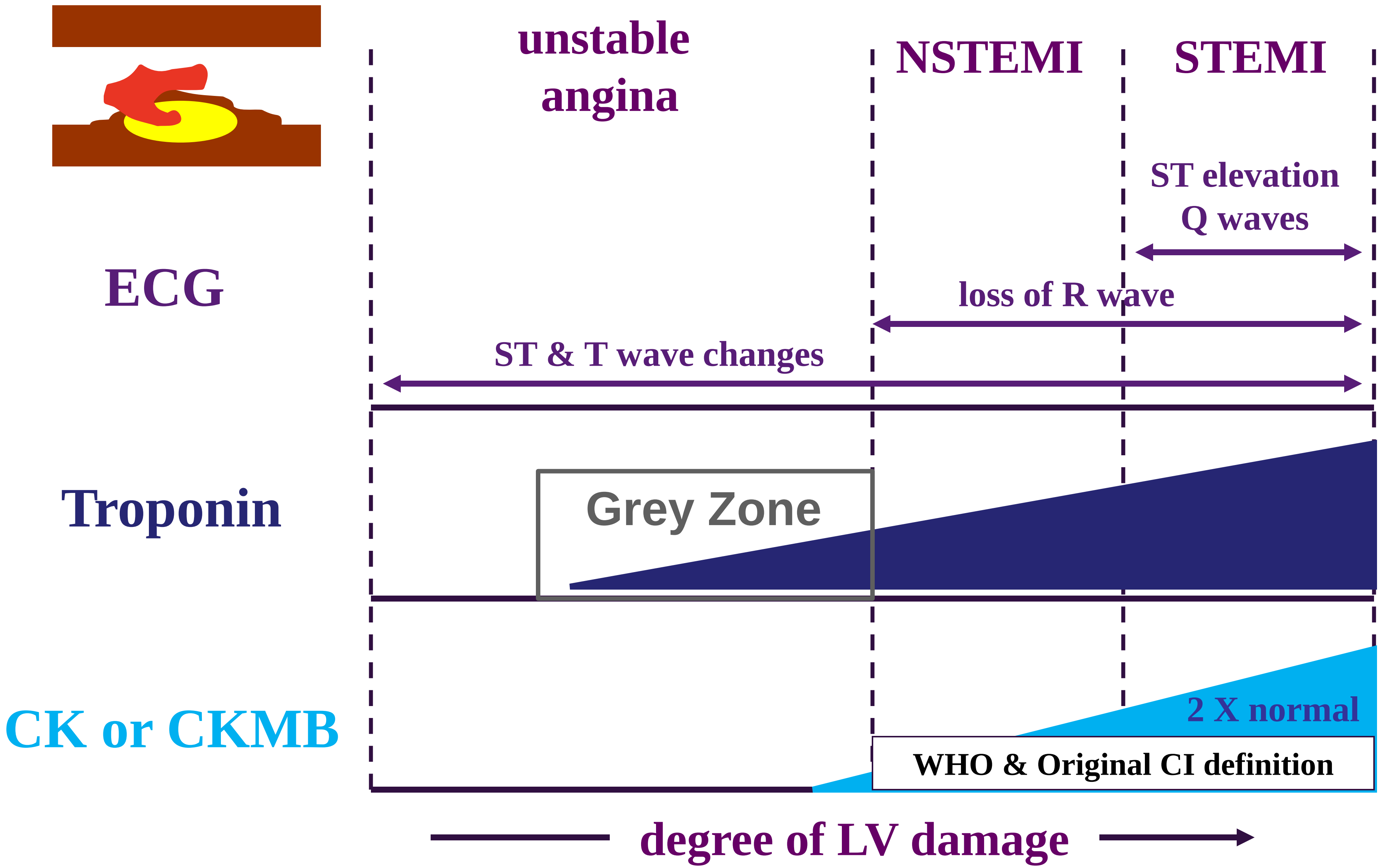
Implications of adopting primary PCI for STEMI

- ♥ **Alters the focus of rehabilitation**

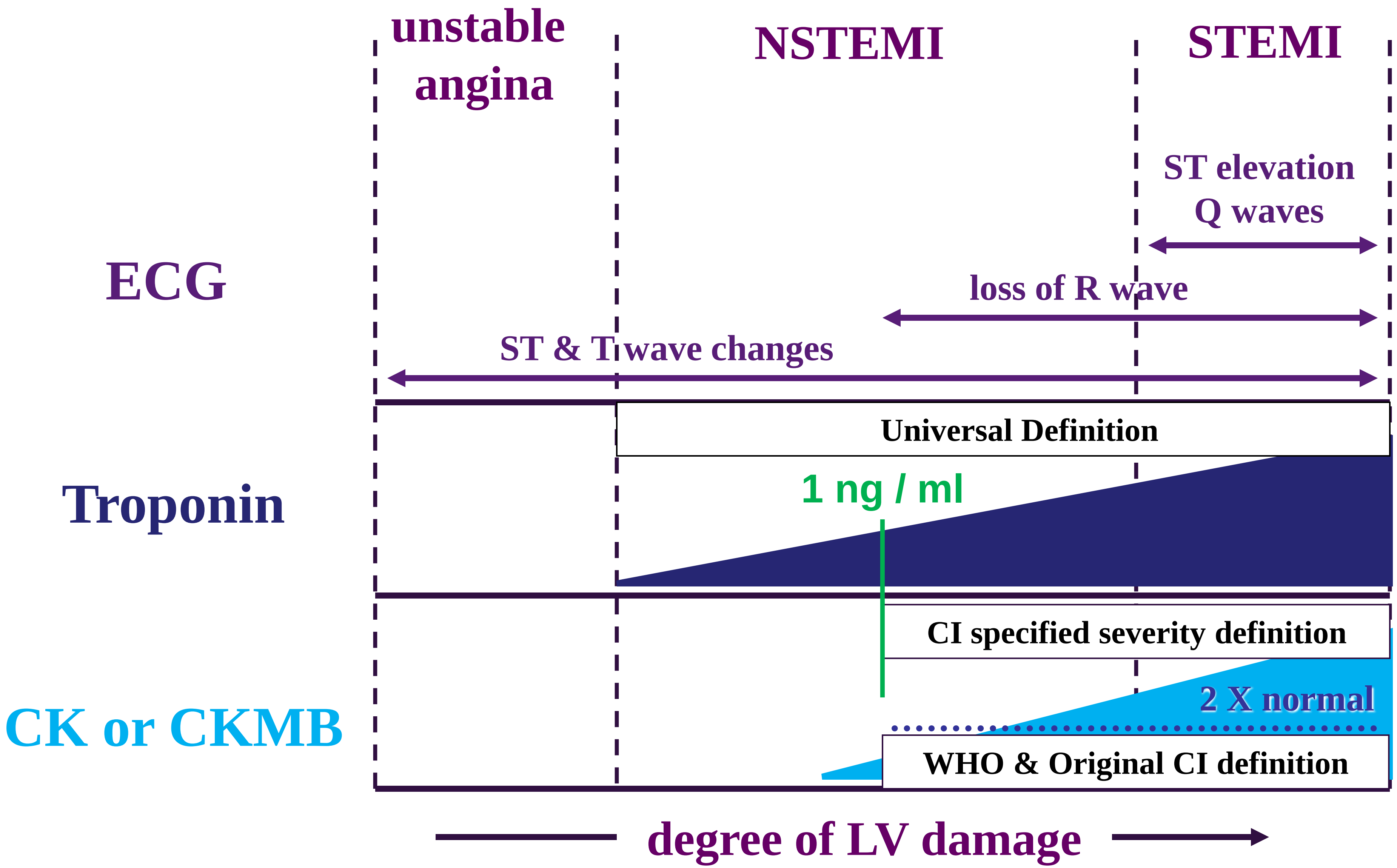
Implications of adopting primary PCI for STEMI

- Alters the focus of rehabilitation
- Modifies troponin release

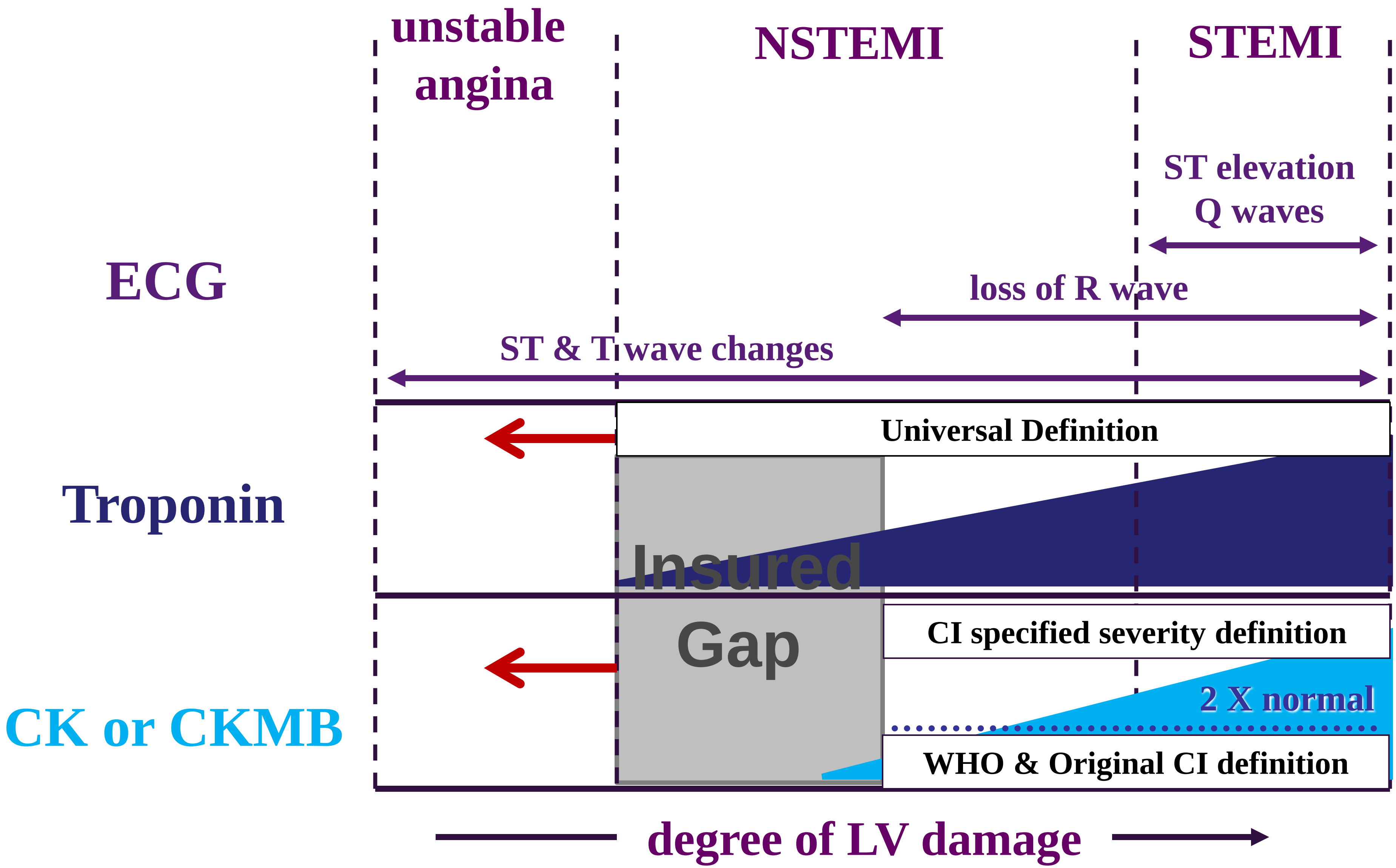
Interpretation of Investigations and Classification of Acute Coronary Syndromes



Interpretation of Investigations and Classification of Acute Coronary Syndromes



Interpretation of Investigations and Classification of Acute Coronary Syndromes



Troponin Testing
lower limit of detection

Ist generation tests (1990)	0.2 ng/ml (ug/L)
HS tests (2008)	0.05 ng/ml
Ultra-sensitive tests (2011)	0.0001 ng/ml

All results should now be reported in ng/L

Real World Experience


Royal Infirmary of Edinburgh 2008

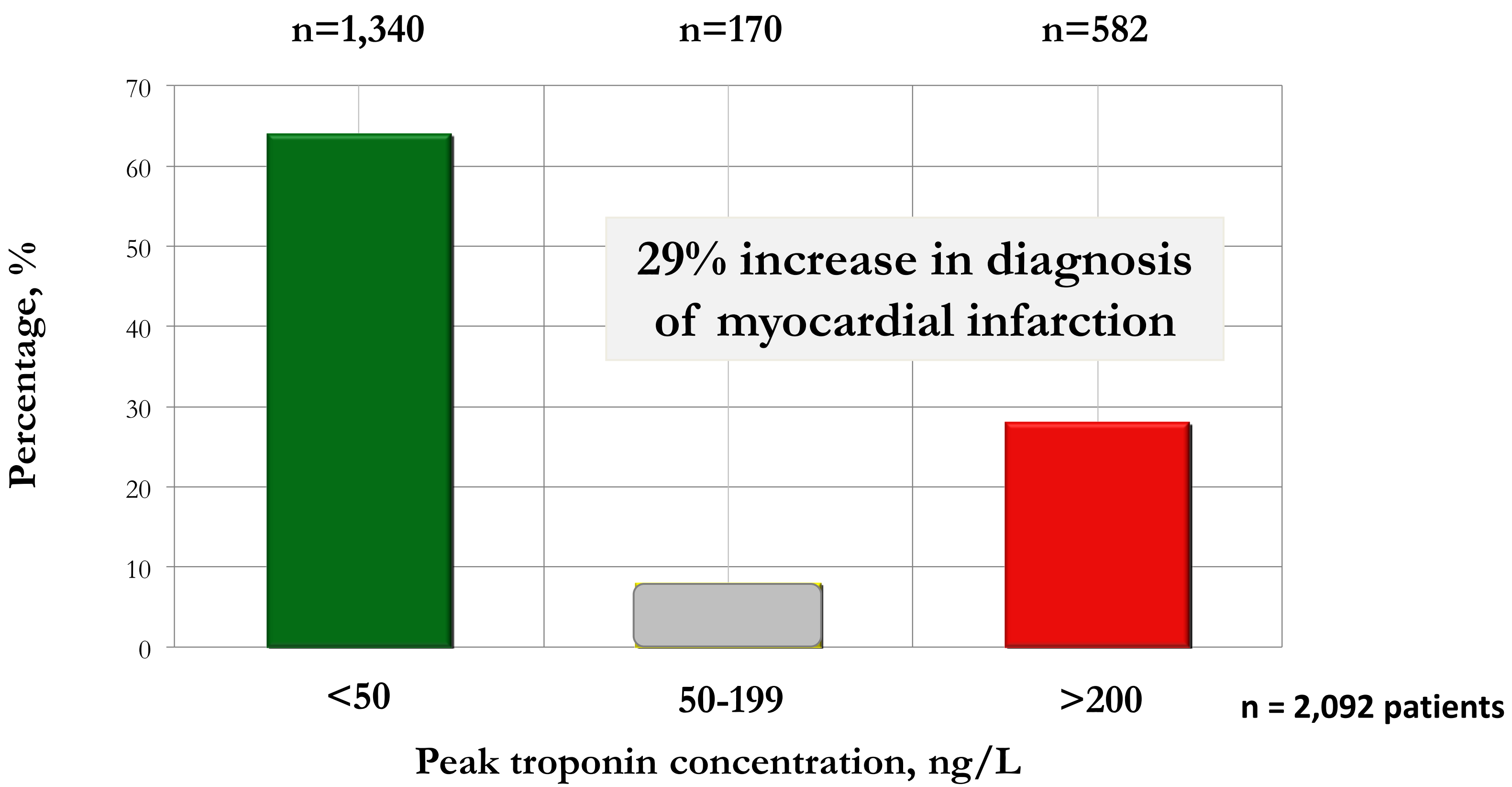
Implementation of a new Sensitive Troponin I Assay (Architect STAT)

Existing Diagnostic Threshold > 200 ng/L

Proposed new Diagnostic Threshold > 50 ng/L

Mills NL et al JAMA. 2011;305(12):1210-1216

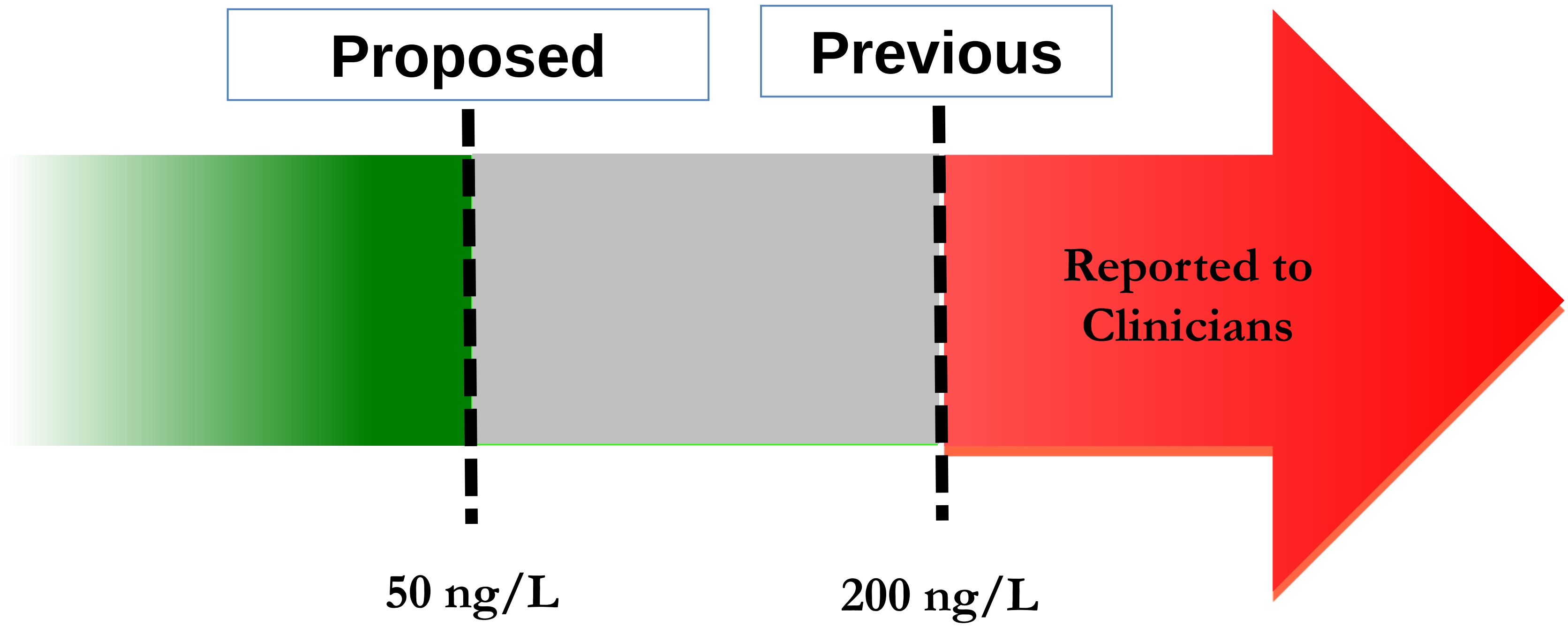
Troponin concentration in consecutive patients with suspected acute coronary syndrome



Estimated extra 626 patients with myocardial infarction in our region
Estimated additional 48,000 patients per annum across UK

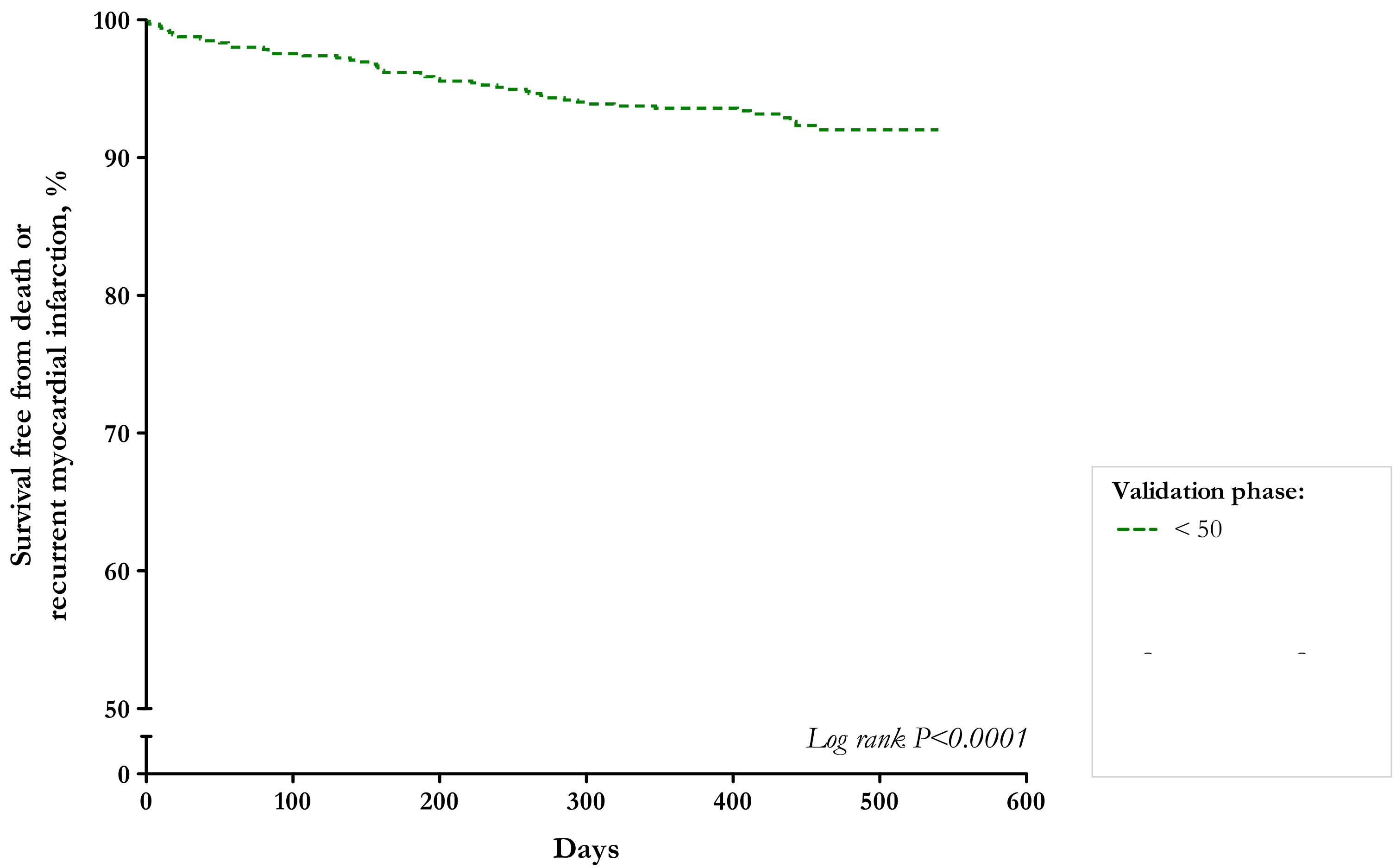
Validation phase 1Feb 2008 – 31 July 2008

Abbott ARCHITECT sensitive troponin I assay



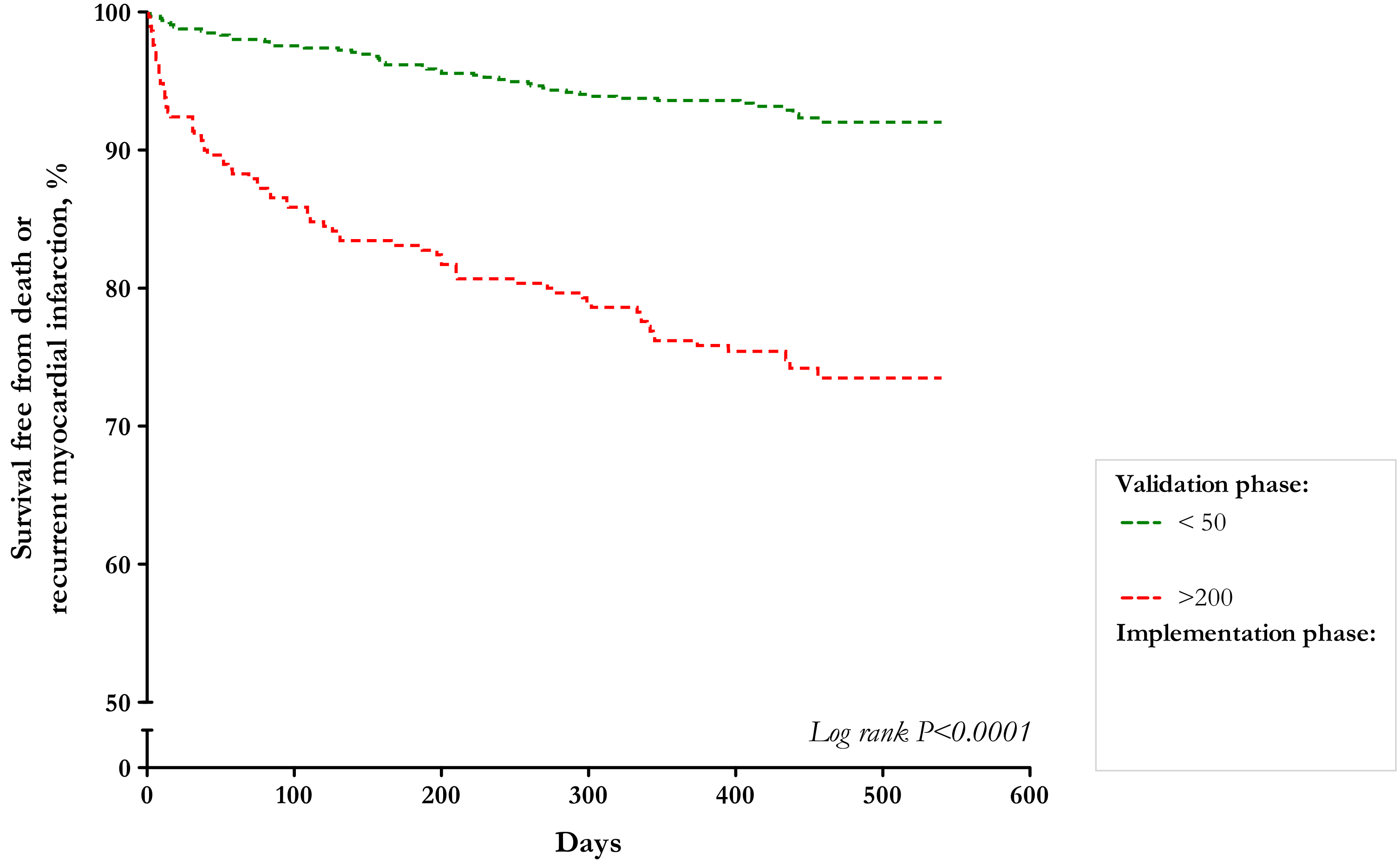
Investigators and clinicians blinded to the results of the sensitive assay below the previous diagnostic threshold (<200 ng/L)

Survival free from death or recurrent myocardial infarction during the validation phase



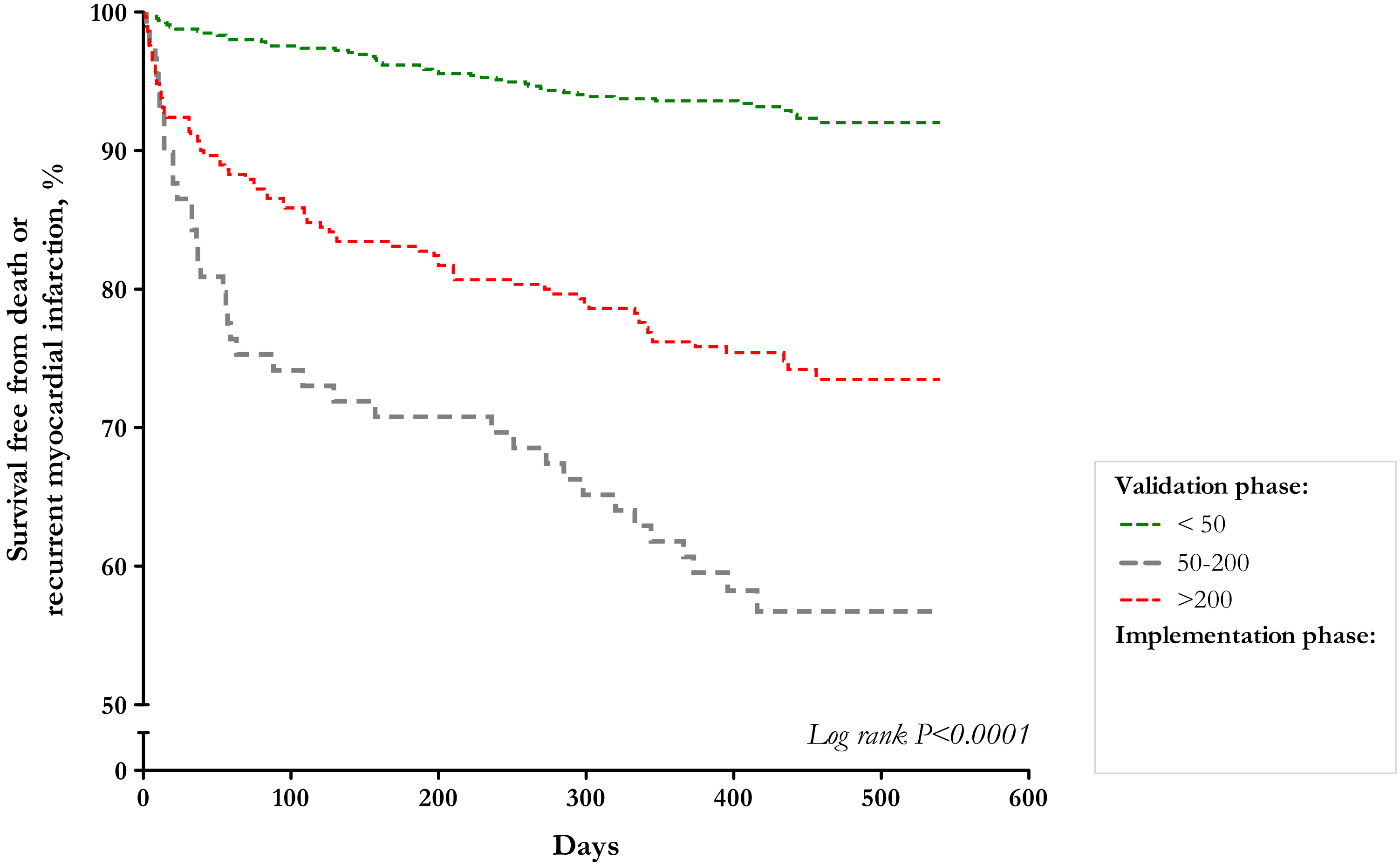
Mills NL et al JAMA. 2011;305(12):1210-1216

Survival free from death or recurrent myocardial infarction during the validation phase



Mills NL et al JAMA. 2011;305(12):1210-1216

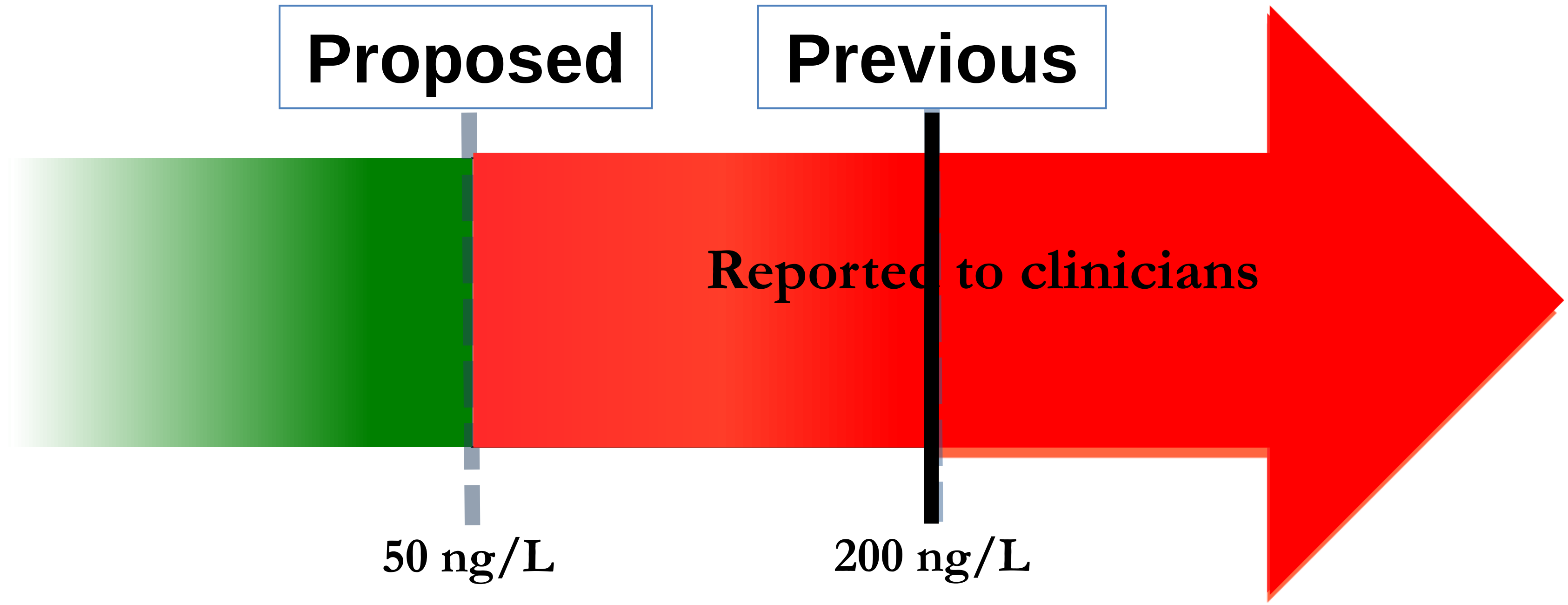
Survival free from death or recurrent myocardial infarction during the validation phase



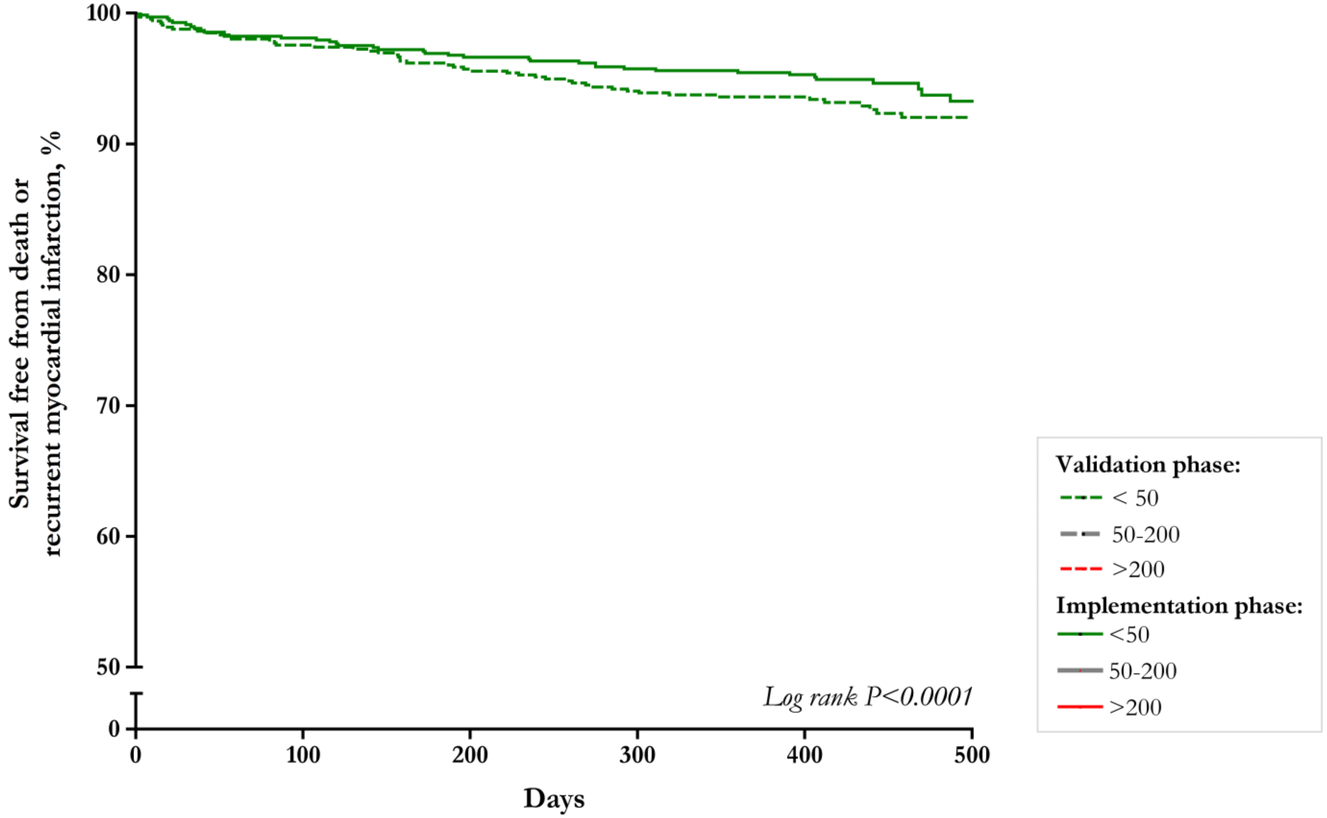
Mills NL et al JAMA. 2011;305(12):1210-1216

Implementation phase 1 Feb 2009 – 31 July 2009

Abbott ARCHITECT sensitive troponin I assay

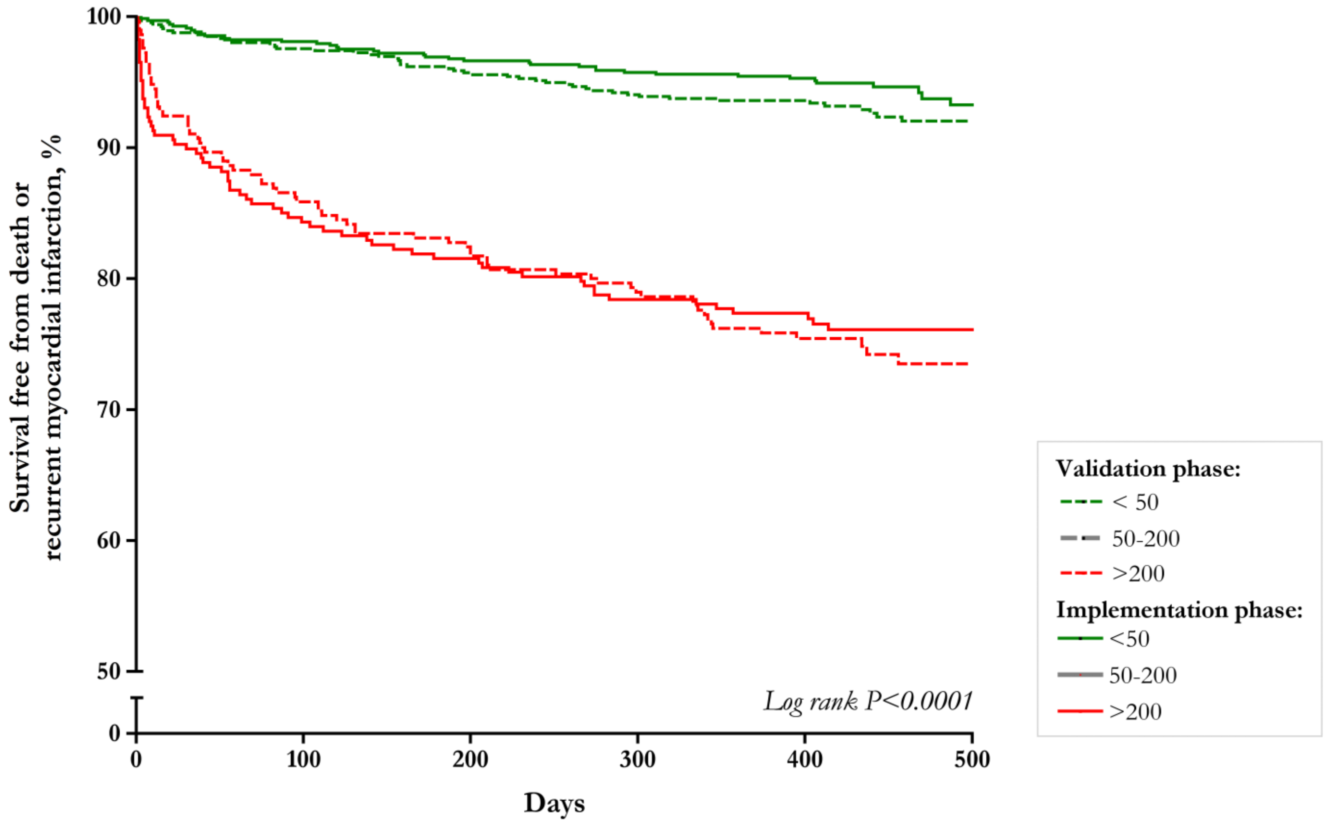


Survival free from death or recurrent myocardial infarction



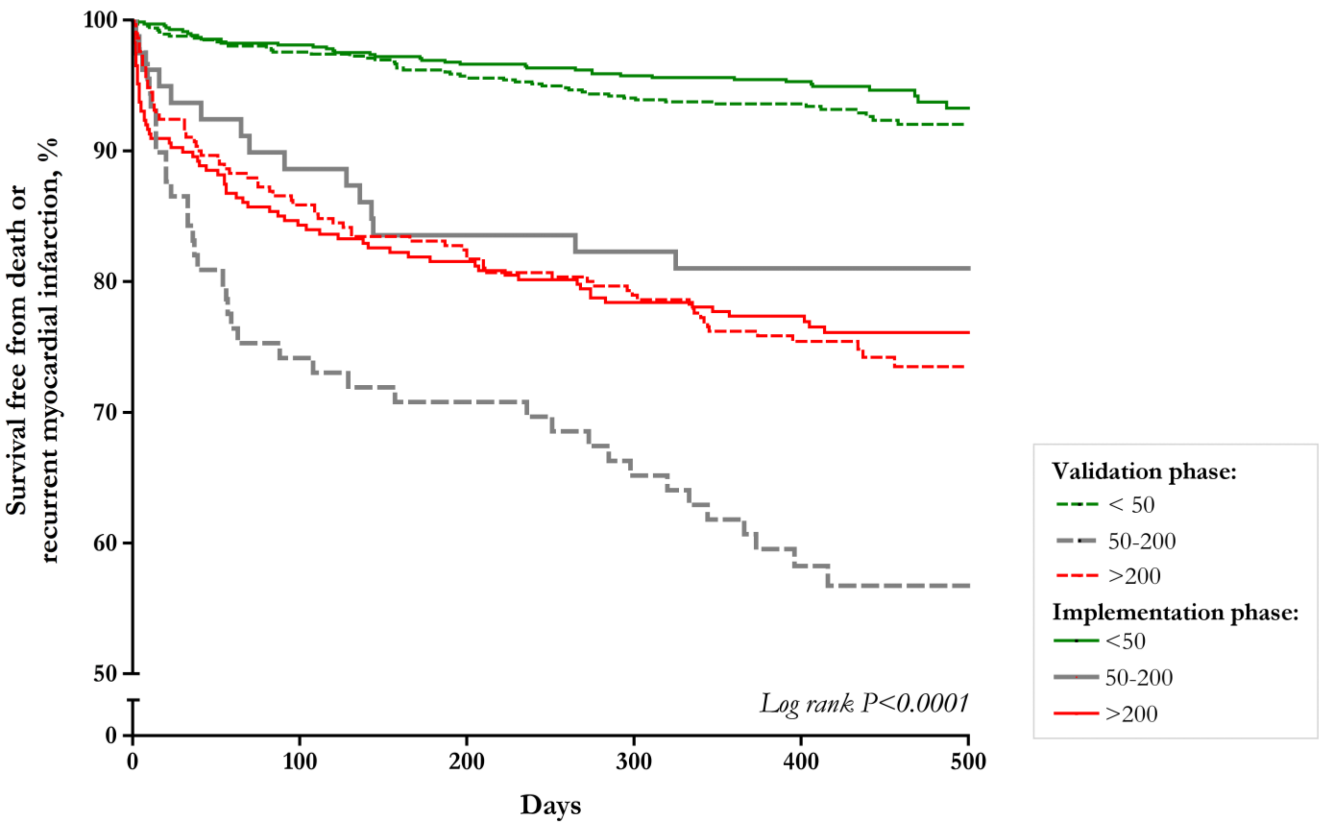
Mills NL et al JAMA. 2011;305(12):1210-1216

Survival free from death or recurrent myocardial infarction



Mills NL et al JAMA. 2011;305(12):1210-1216

Survival free from death or recurrent myocardial infarction



Mills NL et al JAMA. 2011;305(12):1210-1216

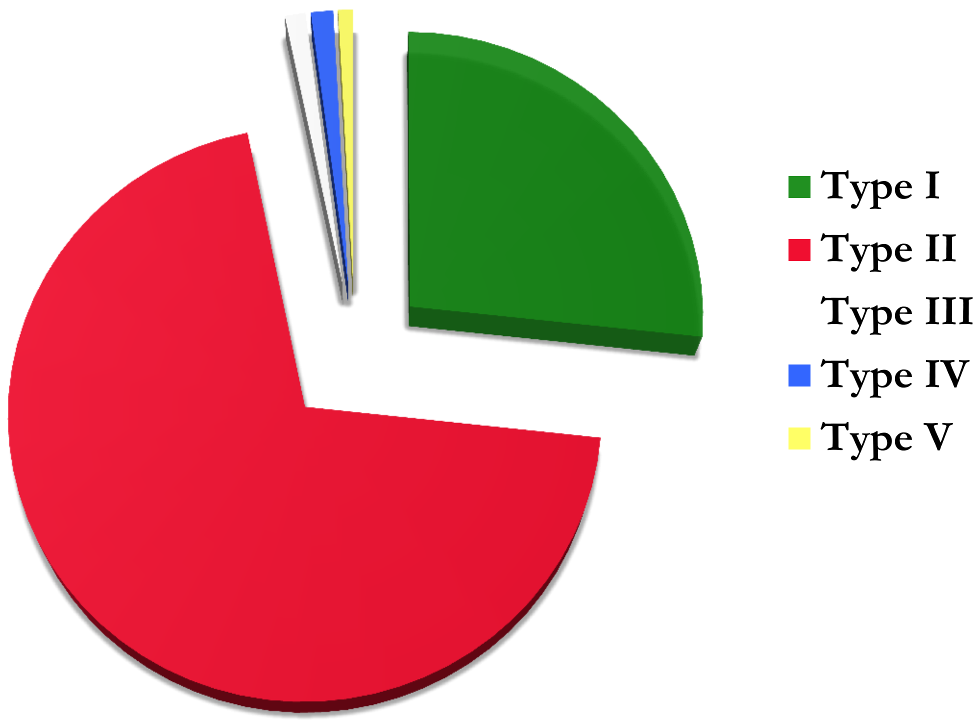
Suspected ACS

Troponin 50 – 200 ng/L

	Validation	Implementation
Specialist Care	**43 %**	**74 %**
DAPT	**28 %**	**58 %**
Angiography	**20 %**	**47 %**

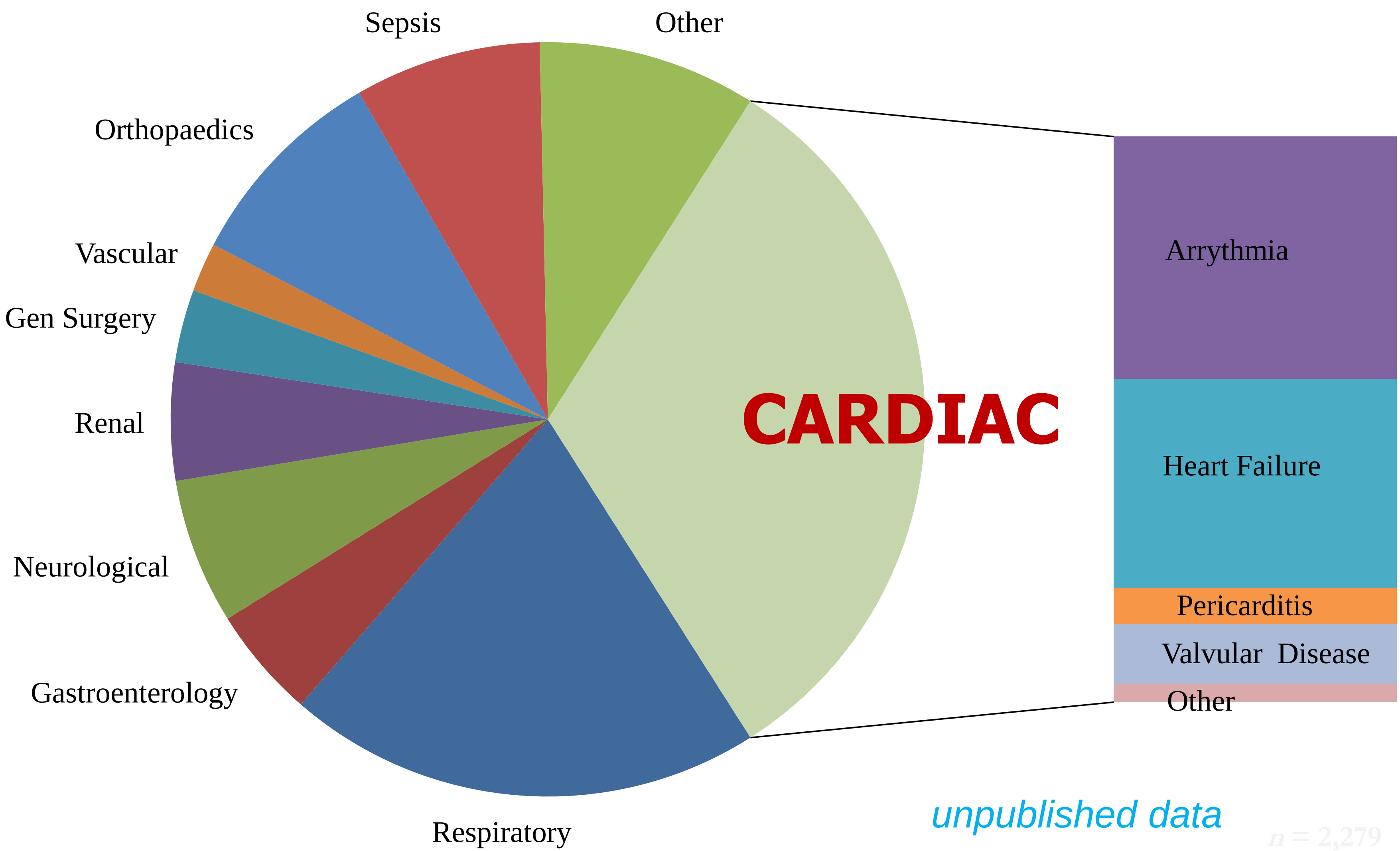
Hospitalised patients classified by type of myocardial infarction

Troponin concentration 50-199 ng/L (n=935)



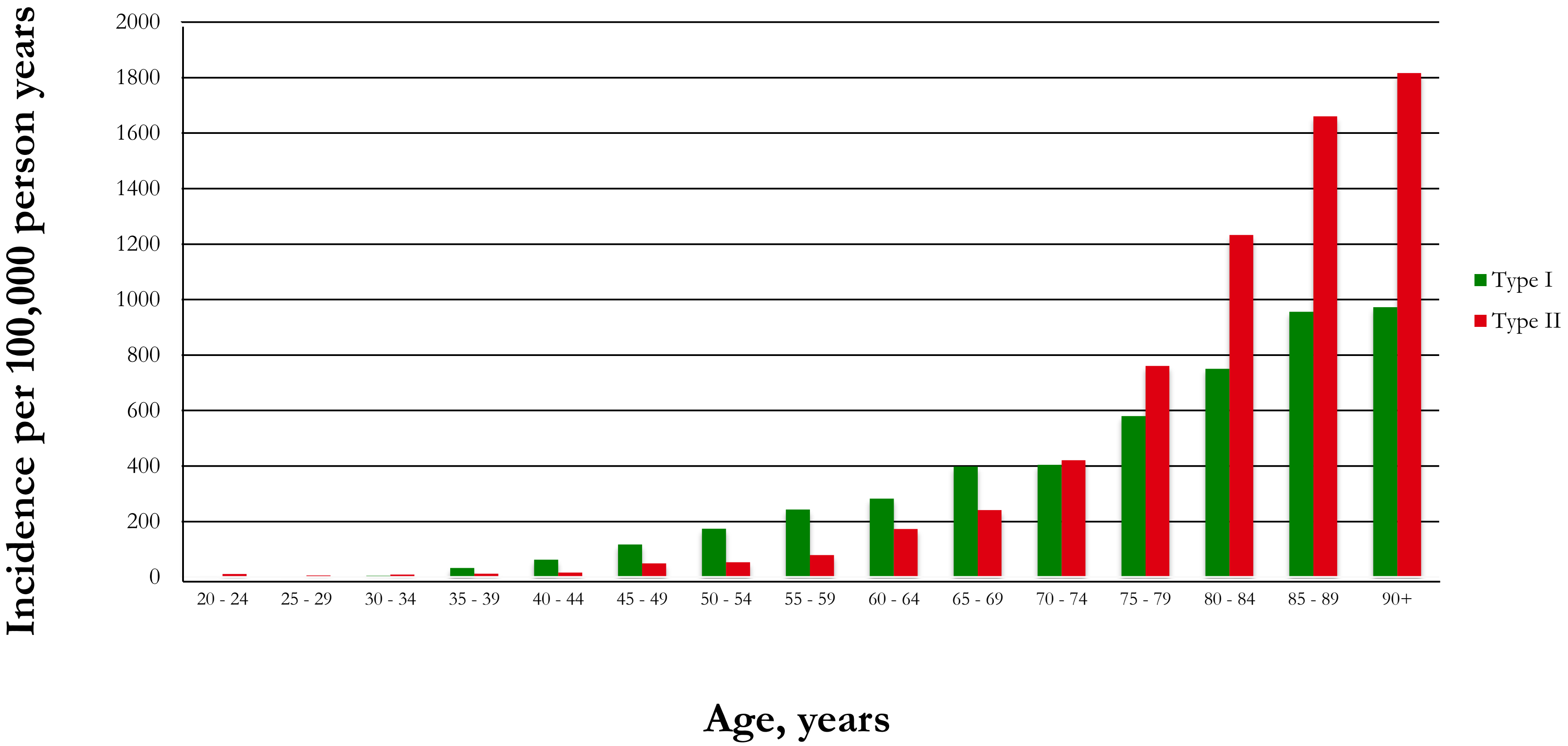
For every patient with type 1 myocardial infarction using a sensitive troponin assay identifies ~3 patients with secondary myocardial ischemia

Primary diagnosis in patients with type 2 myocardial infarction at RIE



Incidence of Type 1 and Type 2 Myocardial Infarction Royal Infirmary of Edinburgh

unpublished data



Detects and improves outcomes in patients with previously undiagnosed type 1 MI

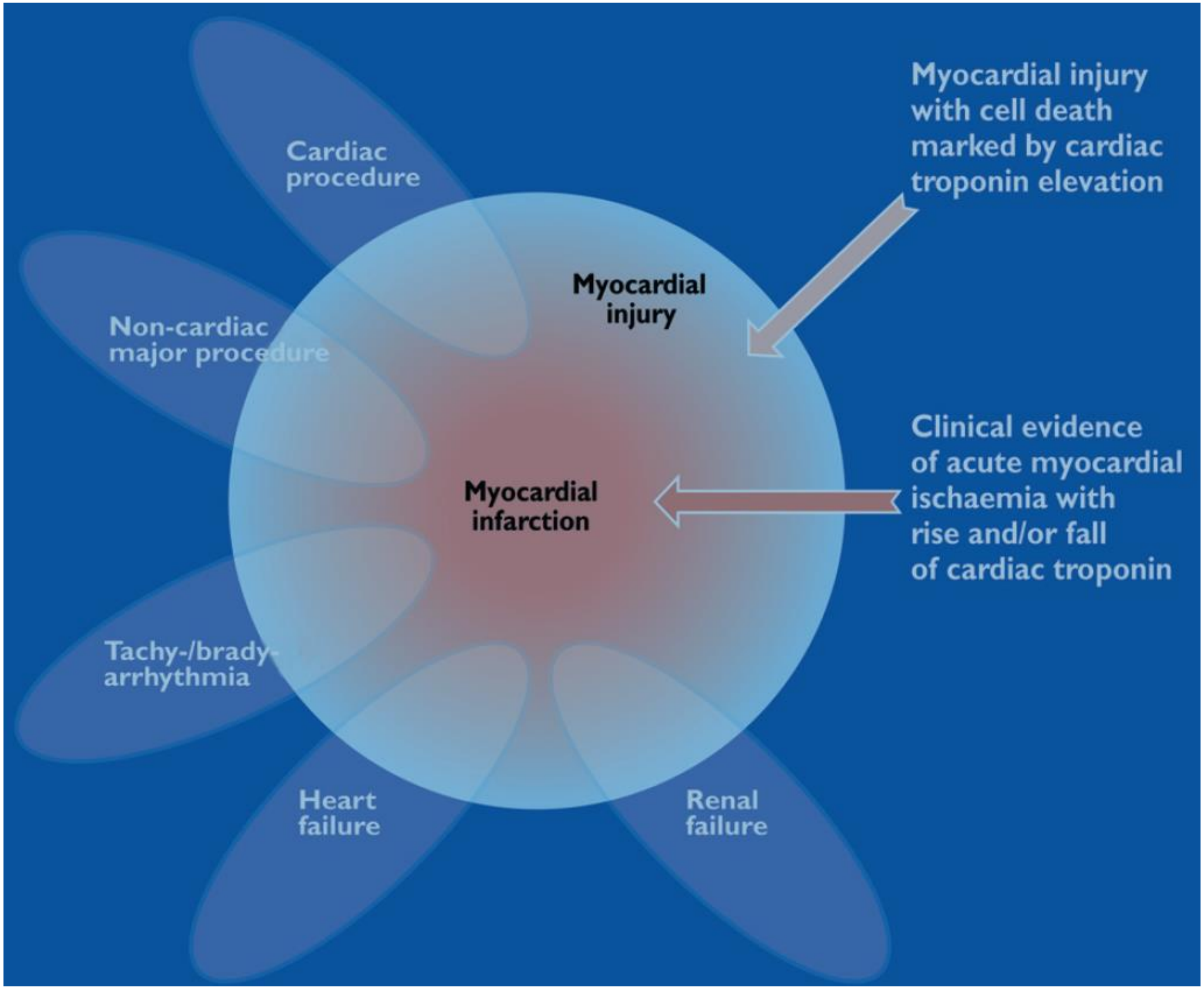
The blessing and the curse of "sensitive" troponin testing

Identifies three times as many patients with other forms of myocardial damage

“When troponin was a lousy assay it was a great test, but now that it’s becoming a great assay, it’s getting to be a lousy test.”

Jesse RL. J AmColl Cardiol. 2010;55:2125–2128.

3rd Universal Definition of MI



http://circ.ahajournals.org/content/early/2012/08/23/CIR.0b013e31826e1058.citation

3rd Universal Definition of Myocardial Infarction

"The term myocardial infarction (MI) should only be used when there is evidence of myocardial necrosis in a clinical setting consistent with myocardial ischaemia"

Detection of a rise and/or fall of cardiac biomarkers with at least one value > 99th percentile of the reference population (where assay CV < 10%) with at least one of the following:

- **symptoms of ischaemia**
- **new ST-T changes or LBBB**
- **development of pathological Q-waves**
- **Imaging evidence of new loss of viable myocardium new regional wall motion abnormality**
- **Identification of an intracoronary thrombus by angiography**

http://circ.ahajournals.org/content/early/2012/08/23/CIR.0b013e31826e1058.citation

3rd Universal Definition of Myocardial Infarction

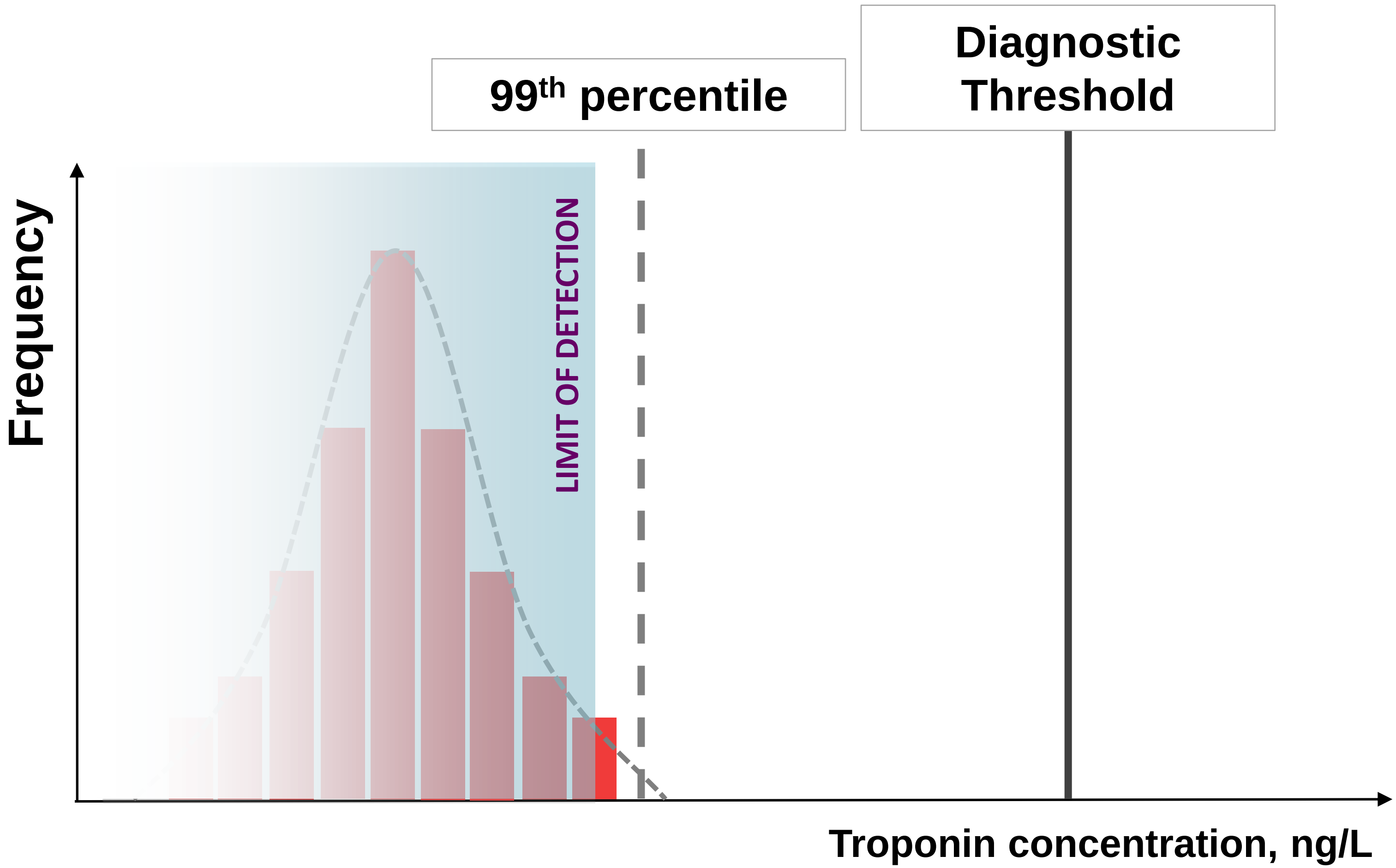
“The term myocardial infarction (MI) should only be used when there is evidence of myocardial necrosis in a clinical setting consistent with myocardial ischaemia”

Detection of a rise and/or fall of cardiac biomarkers with at least one value > 99th percentile of the reference population (where assay CV < 10%) with at least one of the following:

- **symptoms of ischaemia**
- **new ST-T changes or LBBB**
- **development of pathological Q-waves**
- **Imaging evidence of new loss of viable myocardium new regional wall motion abnormality**
- **Identification of an intracoronary thrombus by angiography**

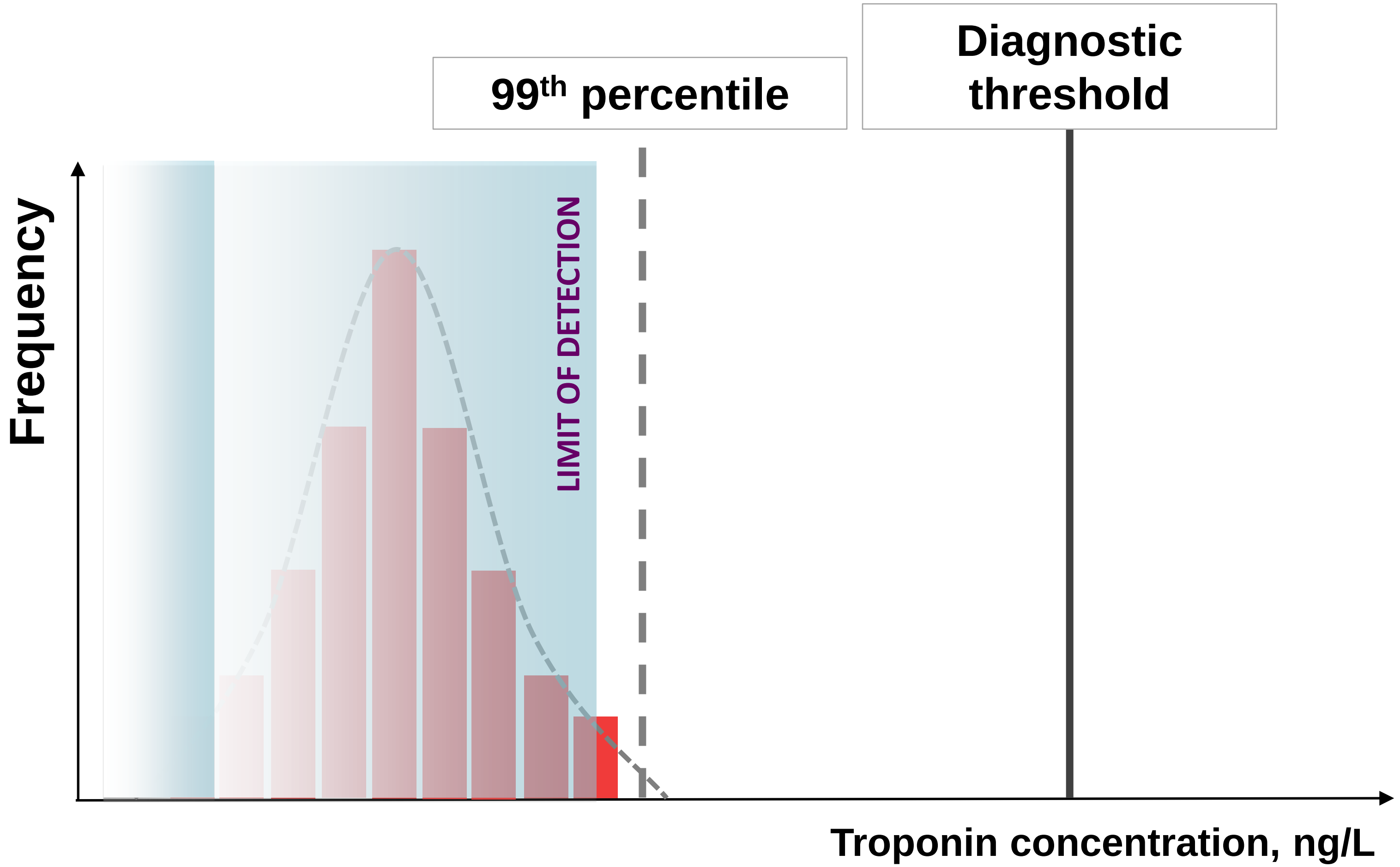
http://circ.ahajournals.org/content/early/2012/08/23/CIR.0b013e31826e1058.citation

Pre 2012 troponin assays



Majority of reference population below limit of detection (>95%) therefore 99th percentile uncertain and diagnostic thresholds based on measure of assay precision (10% CV)

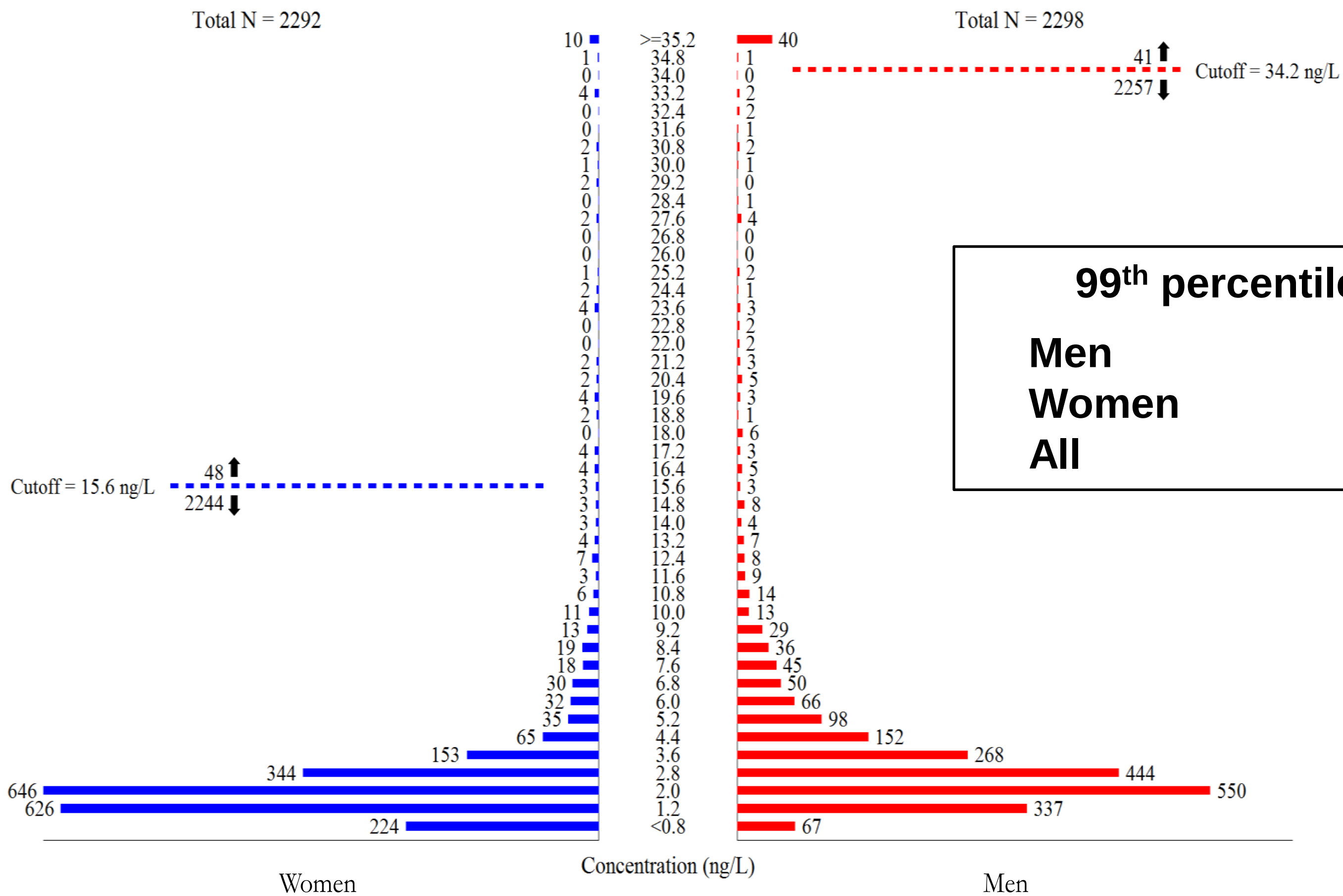
New high-sensitivity troponin assays



High-sensitivity assay have coefficient of variation of <10% at the 99th percentile upper reference limit and troponin detectable in >50% of reference population

ARCHITECT high-sensitive troponin I normal reference range study

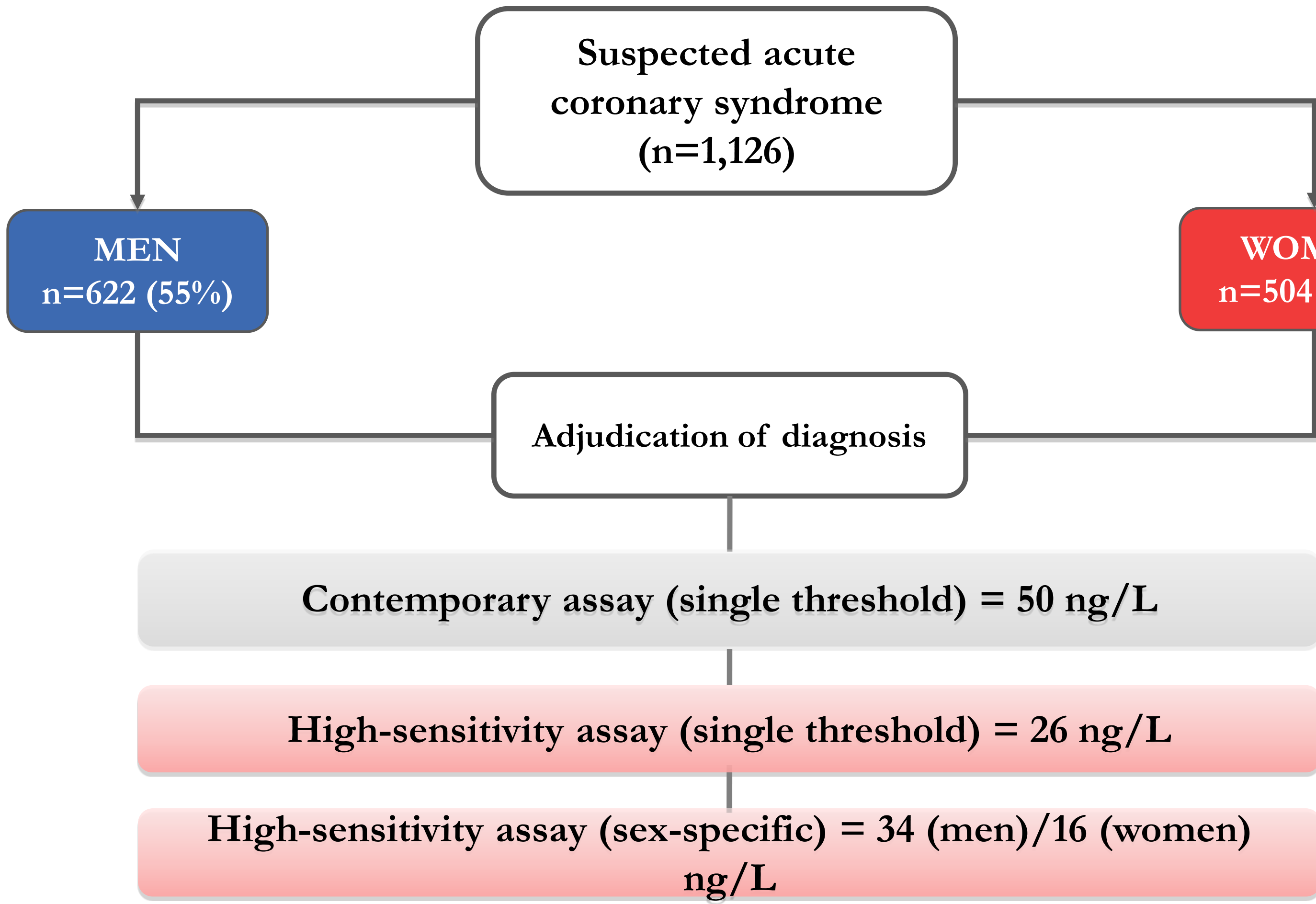
Distribution of troponin and 99th percentile upper reference limits (URL) in 4,590 samples from healthy men and women



99th percentile URL

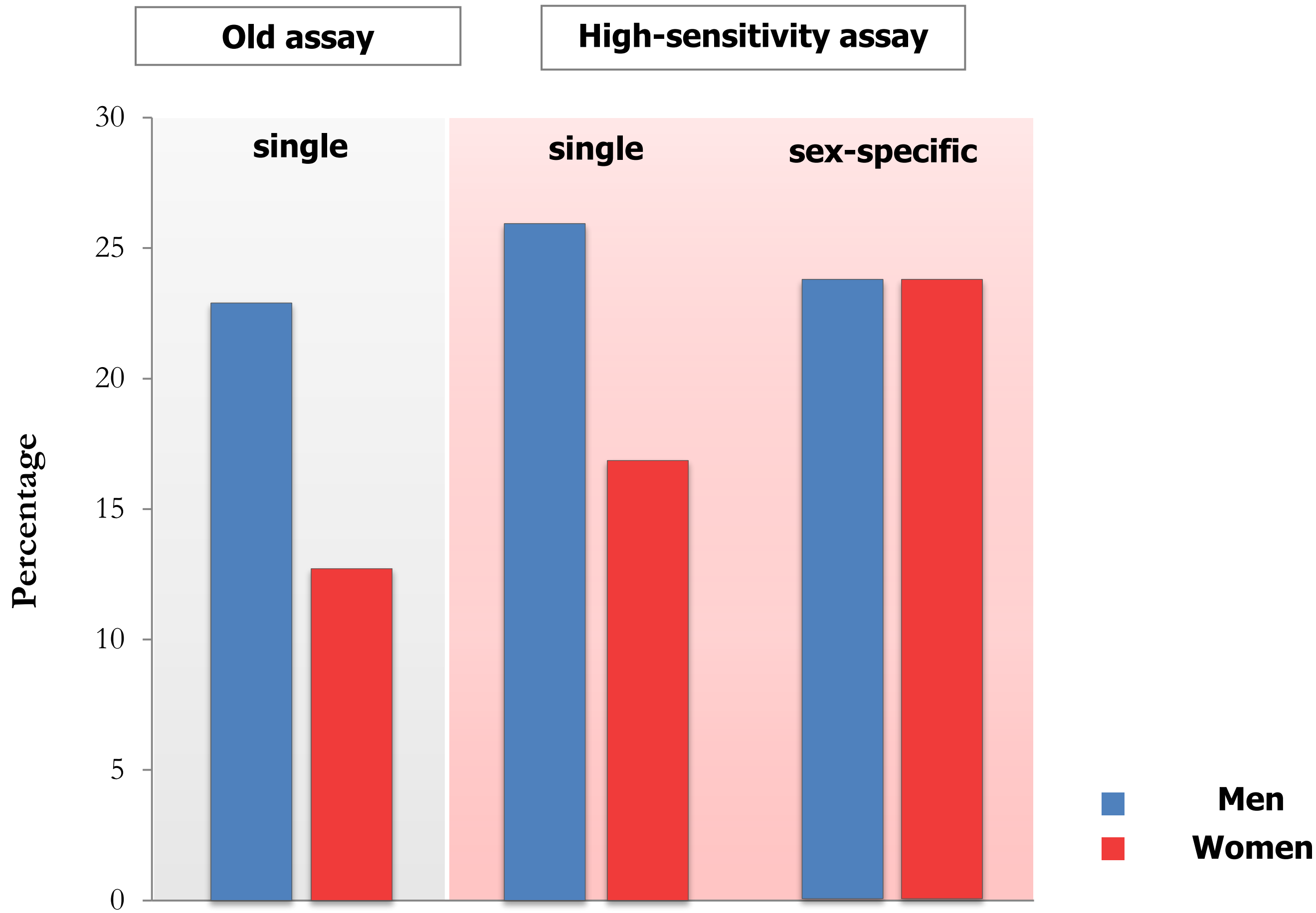
Men	**34 ng/L**
Women	**16 ng/L**
All	**26 ng/L**

Introduction of hs troponin test to RIE 2012



BMJ 2015;350:g7873

Diagnosis of type 1 myocardial infarction



Men	23%	26%	24%
Women	13%	17%	24%

FREQUENCY

BMJ 2015;350:g7873

Summary
Troponin Testing in 2015

<u>High Sensitivity</u> (low diagnostic threshold)

- ♥ More patients are now diagnosed with non ST elevation MI (fewer cases of unstable angina)
- ♥ Useful early triage test
- ♥ Improved clinical outcomes

<u>Low Specificity</u>

- ♥ huge increase in “troponin positive” illnesses that are not due to coronary thrombosis

Made your working lives more difficult (interesting !)